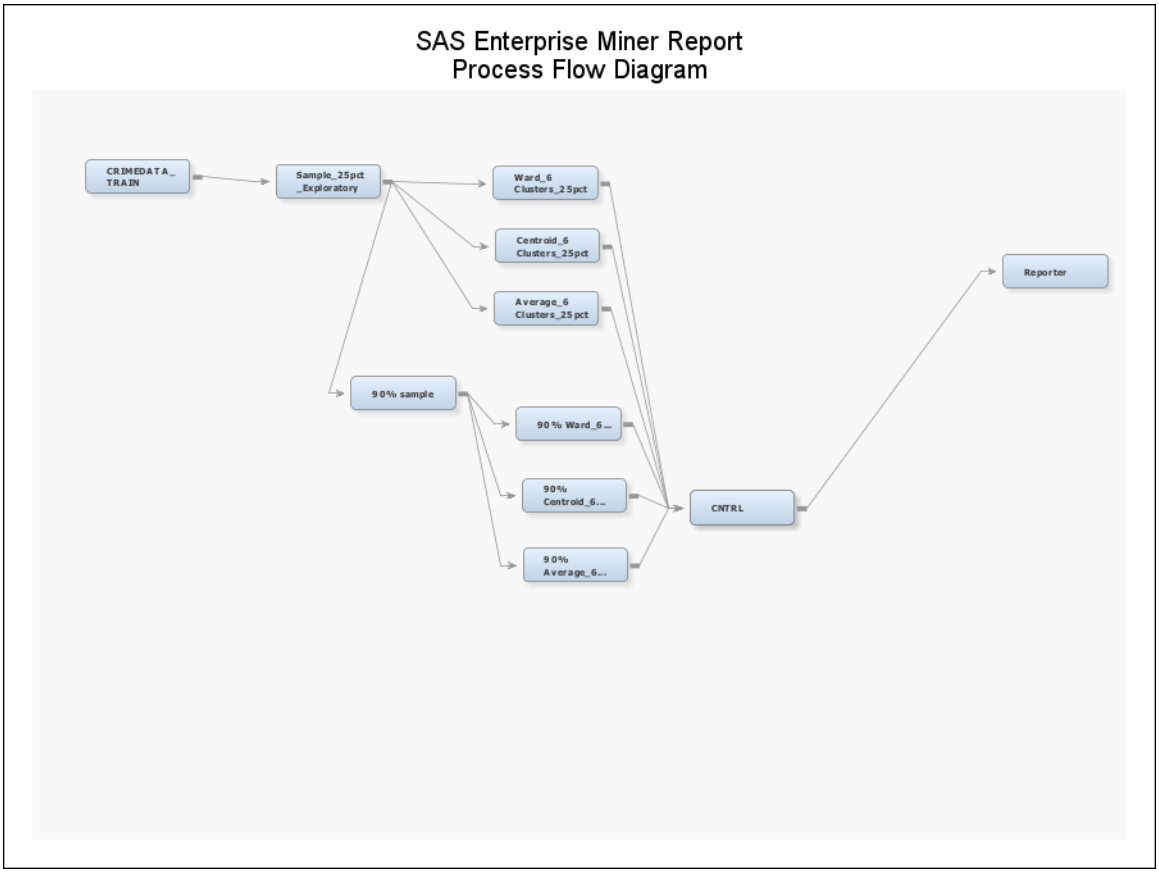
SAS Enterprise Miner Report

User = achas6
Date = 20:03:18 March 04
Project = Chicago Crime 2018_2020
Diagram = ChicagoCrime_Clustering_2018_2020_25pct

Start Node = Report
Node label = Reporter
Nodes = PATH
Showall = N

Format = PDF
Style = LISTING



SAS Enterprise Miner Report

Node=CRIMEDATA_TRAIN
Summary

Node id = Ids
Node label = CRIMEDATA_TRAIN
Meta path = Ids
Notes =

Node=CRIMEDATA_TRAIN
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DataSource		DsCreatedBy	ajay10		NBytes	175440896	.
ApplyIntervalLevelLowerLimit	Y		DsId	crimedatatrain		NCols	15	.
ApplyMaxClassLevels	Y		DsModifiedBy	ajay10		NObs	727944	.
ApplyMaxPercentMissing	Y		DsModifyDate	2087911927.6		NewTable		
CMeta	WORK.M3IDOXFE		DsSampleName			NewVariableRole	REJECT	
ComputeStatistics	N		DsSampleSize			OutputType	VIEW	
DBPassThrough	Y		DsSampleSizeType			Role	RAW	TRAIN
Data	AJDATA.CRIMEDATA_TRAIN		DsScope	LOCAL		Sample	D	
DataSelection	DATASOURCE		IdentifyEmptyColumns	Y		SampleSizeObs	10000	
DataSource	crimedatatrain		IntervalLowerLimit	20		SampleSizePercent	20	
DataSourceRole	RAW		Library	AJDATA		SampleSizeType	PERCENT	
Description			MaxClassLevels	20		Scope	LOCAL	
DropMapVariables	Y		MaxPercentMissing	50		Segment		
DsCreateDate	2087911927.6		MetaAdvisor	BASIC		Table	CRIMEDATA_TRAIN	

Node=CRIMEDATA_TRAIN
Data Attributes

Attribute	Value	Attribute	Value	Attribute	Value
Data Name	CRIMEDATA_TRAIN	Date Created	27Feb2026:22:28:16	Data Size	175440896
Data Type	DATA	Date Modified	27Feb2026:22:28:16	Role	RAW
Data Label		Number Rows	727944	Segment	
Engine	BASE	Number Columns	15	Data Library	AJDATA

Node=CRIMEDATA_TRAIN
Variables List

Name	Label	Role	Level	Type	Length	Format	Creator
Arrest		INPUT	BINARY	N	8	BEST12.0	
Beat		REJECTED	NOMINAL	N	8	BEST12.0	
Community_Area	Community Area	INPUT	NOMINAL	N	8		
Description		REJECTED	NOMINAL	C	60	$60.	
District		INPUT	NOMINAL	N	8	BEST12.0	
Domestic		INPUT	BINARY	N	8	BEST12.0	
Hour		INPUT	INTERVAL	N	8	BEST12.0	
ID		REJECTED	NOMINAL	N	8	BEST12.0	
Latitude		INPUT	INTERVAL	N	8	BEST12.0	
Location_Description	Location Description	REJECTED	NOMINAL	C	55		
Longitude		INPUT	INTERVAL	N	8	BEST12.0	

Name	Label	Role	Level	Type	Length	Format	Creator
Month		INPUT	INTERVAL	N	8	BEST12.0	
Primary_Type	Primary Type	REJECTED	NOMINAL	C	26		
Ward		INPUT	NOMINAL	N	8	BEST12.0	
Year		REJECTED	INTERVAL	N	8	BEST12.0	

SAS Enterprise Miner Report

Node=Sample_25pct_Exploratory
Summary

Node id = Smpl
Node label = Sample_25pct_Exploratory
Meta path = Ids => Smpl
Notes =

Node=Sample_25pct_Exploratory
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Sample		IgnoreSmallStrata	N		OutputType	DATA	
AdjustFreq	N		IntervalDistribution	Y		Pvalue	0.01	
Alpha	0.01		LevelProportion	100		RandomSeed	12345	
ClassDistribution	Y		LevelSampleProportion	50		SizeObs	.	
ClusterMethod	RANDOM		LevelSelection	EVENT		SizePercent	25	10
FreqCount	N		Method	DEFAULT		SizeType	PERCENT	
FreqMiss	N		MinStrataSize	5		StratifyCriterion	PROPORTIONAL	

Node=Sample_25pct_Exploratory
Variable Summary

Role	Level	Frequency Count	Name
REJECTED	INTERVAL	1	Year
REJECTED	NOMINAL	5	Beat Description ID Location_Description Primary_Type
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward

SAS Enterprise Miner Report

Node=90% sample
Summary

Node id = Smpl2
Node label = 90% sample
Meta path = Ids => Smpl => Smpl2
Notes =

Node=90% sample
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Sample		IgnoreSmallStrata	N		OutputType	DATA	
AdjustFreq	N		IntervalDistribution	Y		Pvalue	0.01	
Alpha	0.01		LevelProportion	100		RandomSeed	12345	
ClassDistribution	Y		LevelSampleProportion	50		SizeObs	.	
ClusterMethod	RANDOM		LevelSelection	EVENT		SizePercent	90	10
FreqCount	N		Method	DEFAULT		SizeType	PERCENT	
FreqMiss	N		MinStrataSize	5		StratifyCriterion	PROPORTIONAL	

Node=90% sample
Variable Summary

Role	Level	Frequency Count	Name
REJECTED	INTERVAL	1	Year
REJECTED	NOMINAL	5	Beat Description ID Location_Description Primary_Type
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

SAS Enterprise Miner Report

Node=Average_6Clusters_25pct
Summary

Node id = Clus5
Node label = Average_6Clusters_25pct
Meta path = Ids => Smpl => Clus5
Notes =

Node=Average_6Clusters_25pct
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	AVERAGE	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=Average_6Clusters_25pct
Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=Average_6Clusters_25pct
Mean Statistics

Variable	Segment 1	Segment 2
Clustering Criterion	0.24	0.2381
Maximum Relative Change in Cluster Seeds	0.00	0.0000
Improvement in Clustering Criterion	0.00	0.0000
Frequency of Cluster	181980.00	6.0000
Root-Mean-Square Standard Deviation	0.24	0.2091
Maximum Distance from Cluster Seed	6.04	3.3382
Nearest Cluster	2.00	1.0000
Distance to Nearest Cluster	36.54	36.5434
Hour	13.01	9.8333
Latitude	41.84	41.9809
Longitude	-87.67	-87.8618
Month	6.55	8.0000

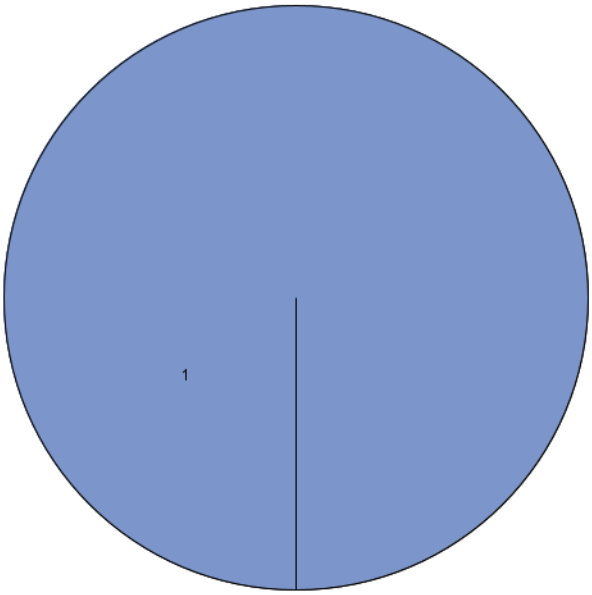
Variable	Segment 1	Segment 2
Arrest=0	0.80	0.8333
Arrest=1	0.20	0.1667
Community_Area=1	0.01	0.0000
Community_Area=2	0.01	0.0000
Community_Area=3	0.01	-0.0000
Community_Area=4	0.01	0.0000
Community_Area=5	0.00	-0.0000
Community_Area=6	0.02	0.0000
Community_Area=7	0.02	0.0000
Community_Area=8	0.04	-0.0000
Community_Area=9	0.00	0.1667
Community_Area=10	0.00	-0.0000
Community_Area=11	0.00	-0.0000
Community_Area=12	0.00	0.0000
Community_Area=13	0.00	0.0000
Community_Area=14	0.01	0.0000
Community_Area=15	0.01	0.0000
Community_Area=16	0.01	0.0000
Community_Area=17	0.01	0.0000
Community_Area=18	0.00	-0.0000
Community_Area=19	0.02	-0.0000
Community_Area=20	0.01	0.0000
Community_Area=21	0.01	0.0000
Community_Area=22	0.02	0.0000
Community_Area=23	0.03	-0.0000
Community_Area=24	0.03	0.1667
Community_Area=25	0.06	0.0000
Community_Area=26	0.02	0.0000
Community_Area=27	0.02	0.0000
Community_Area=28	0.04	0.0000
Community_Area=29	0.04	0.0000
Community_Area=30	0.02	0.0000
Community_Area=31	0.01	0.0000
Community_Area=32	0.04	0.0000
Community_Area=33	0.01	0.0000
Community_Area=34	0.00	0.0000
Community_Area=35	0.01	-0.0000
Community_Area=36	0.00	0.0000
Community_Area=37	0.00	0.0000
Community_Area=38	0.01	0.0000
Community_Area=39	0.01	0.0000
Community_Area=40	0.01	-0.0000
Community_Area=41	0.01	0.0000
Community_Area=42	0.01	-0.0000
Community_Area=43	0.03	0.0000
Community_Area=44	0.03	-0.0000
Community_Area=45	0.01	0.0000
Community_Area=46	0.02	0.0000
Community_Area=47	0.00	-0.0000
Community_Area=48	0.01	0.0000
Community_Area=49	0.03	0.0000

Variable	Segment 1	Segment 2
Community_Area=50	0.00	0.0000
Community_Area=51	0.01	0.0000
Community_Area=52	0.00	0.0000
Community_Area=53	0.02	0.0000
Community_Area=54	0.01	0.0000
Community_Area=55	0.00	0.0000
Community_Area=56	0.01	0.0000
Community_Area=57	0.00	-0.0000
Community_Area=58	0.01	0.0000
Community_Area=59	0.00	0.0000
Community_Area=60	0.01	0.0000
Community_Area=61	0.02	0.0000
Community_Area=62	0.00	0.0000
Community_Area=63	0.01	0.0000
Community_Area=64	0.00	0.0000
Community_Area=65	0.01	0.0000
Community_Area=66	0.02	0.0000
Community_Area=67	0.03	0.0000
Community_Area=68	0.02	0.0000
Community_Area=69	0.03	0.0000
Community_Area=70	0.01	0.0000
Community_Area=71	0.03	-0.0000
Community_Area=72	0.00	0.0000
Community_Area=73	0.01	0.0000
Community_Area=74	0.00	0.0000
Community_Area=75	0.01	-0.0000
Community_Area=76	0.01	0.6667
Community_Area=77	0.01	-0.0000
District=1	0.05	0.0000
District=2	0.04	0.0000
District=3	0.05	-0.0000
District=4	0.05	0.0000
District=5	0.05	0.0000
District=6	0.06	0.0000
District=7	0.05	0.0000
District=8	0.06	-0.0000
District=9	0.04	-0.0000
District=10	0.05	0.0000
District=11	0.07	0.0000
District=12	0.05	0.0000
District=14	0.03	0.0000
District=15	0.04	0.0000
District=16	0.03	0.0000
District=17	0.03	0.0000
District=18	0.05	-0.0000
District=19	0.04	0.0000
District=20	0.02	0.0000
District=22	0.03	0.0000
District=24	0.03	-0.0000
District=25	0.05	0.0000
District=31	-0.00	1.0000

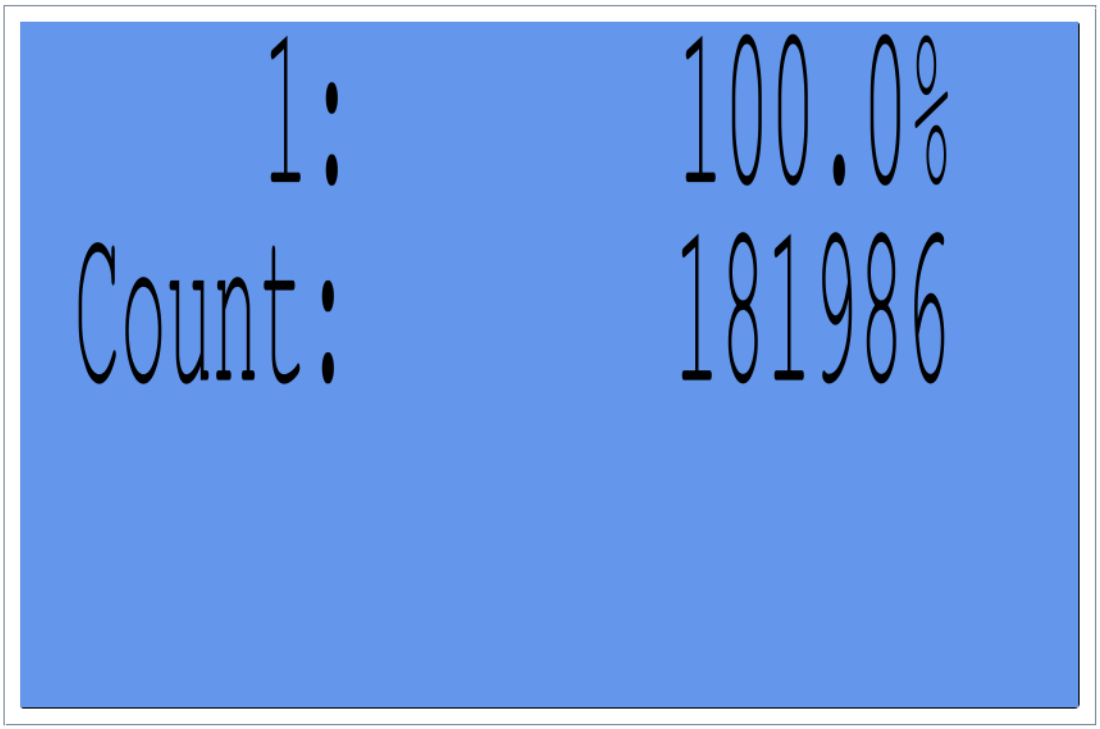
Variable	Segment 1	Segment 2
Domestic=0	0.80	1.0000
Domestic=1	0.20	-0.0000
Ward=1	0.02	0.1667
Ward=2	0.03	0.0000
Ward=3	0.03	0.0000
Ward=4	0.02	0.0000
Ward=5	0.02	-0.0000
Ward=6	0.04	0.0000
Ward=7	0.03	0.0000
Ward=8	0.03	0.0000
Ward=9	0.03	0.0000
Ward=10	0.01	0.0000
Ward=11	0.01	0.0000
Ward=12	0.01	-0.0000
Ward=13	0.01	0.0000
Ward=14	0.01	0.0000
Ward=15	0.02	0.0000
Ward=16	0.03	-0.0000
Ward=17	0.03	0.0000
Ward=18	0.01	-0.0000
Ward=19	0.01	0.0000
Ward=20	0.03	0.0000
Ward=21	0.03	0.0000
Ward=22	0.01	-0.0000
Ward=23	0.01	0.0000
Ward=24	0.04	0.0000
Ward=25	0.02	-0.0000
Ward=26	0.02	-0.0000
Ward=27	0.04	0.0000
Ward=28	0.05	0.0000
Ward=29	0.02	0.0000
Ward=30	0.01	0.0000
Ward=31	0.01	-0.0000
Ward=32	0.01	0.0000
Ward=33	0.01	0.0000
Ward=34	0.03	0.0000
Ward=35	0.01	0.0000
Ward=36	0.01	0.0000
Ward=37	0.03	0.0000
Ward=38	0.01	0.0000
Ward=39	0.01	0.0000
Ward=40	0.01	0.0000
Ward=41	0.01	0.8333
Ward=42	0.06	-0.0000
Ward=43	0.01	0.0000
Ward=44	0.01	0.0000
Ward=45	0.01	-0.0000
Ward=46	0.01	0.0000
Ward=47	0.01	0.0000
Ward=48	0.01	-0.0000

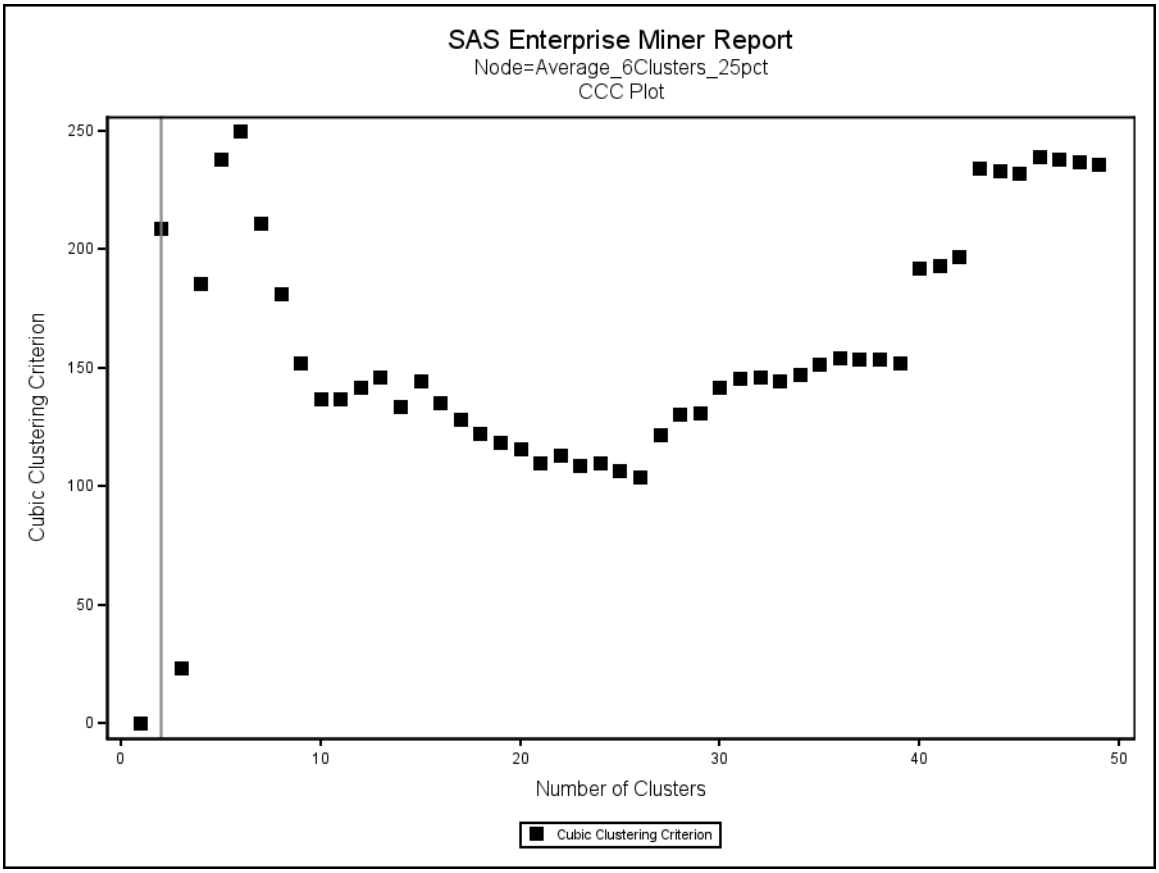
Variable	Segment 1	Segment 2
Ward=49	0.01	-0.0000
Ward=50	0.01	0.0000

SAS Enterprise Miner Report
Node=Average_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=Average_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=Average_6Clusters_25pct
CCC Plot
Cubic Clustering Criterion
250
200
150
100
50
0
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=90% Average_6Clusters_25pct Summary

Node id = Clus6
Node label = 90% Average_6Clusters_25pct
Meta path = Ids => Smpl => Smpl2 => Clus6
Notes =

Node=90% Average_6Clusters_25pct Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	AVERAGE	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=90% Average_6Clusters_25pct Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=90% Average_6Clusters_25pct Mean Statistics

Variable	Segment 1	Segment 2
Clustering Criterion	0.24	0.2381
Maximum Relative Change in Cluster Seeds	0.00	0.0000
Improvement in Clustering Criterion	0.00	0.0000
Frequency of Cluster	163781.00	6.0000
Root-Mean-Square Standard Deviation	0.24	0.2088
Maximum Distance from Cluster Seed	6.03	3.3346
Nearest Cluster	2.00	1.0000
Distance to Nearest Cluster	34.69	34.6917
Hour	13.01	9.8333
Latitude	41.84	41.9809
Longitude	-87.67	-87.8618
Month	6.54	8.0000

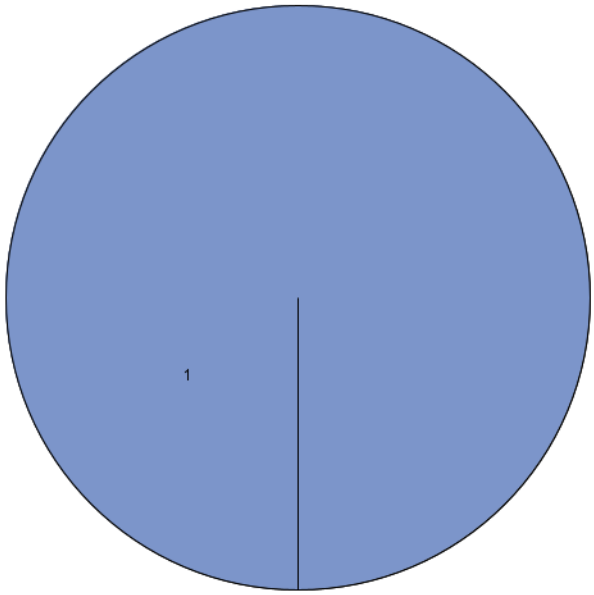
Variable	Segment 1	Segment 2
Arrest=0	0.80	0.8333
Arrest=1	0.20	0.1667
Community_Area=1	0.01	0.0000
Community_Area=2	0.01	0.0000
Community_Area=3	0.01	0.0000
Community_Area=4	0.01	0.0000
Community_Area=5	0.00	0.0000
Community_Area=6	0.02	0.0000
Community_Area=7	0.02	0.0000
Community_Area=8	0.04	-0.0000
Community_Area=9	0.00	0.1667
Community_Area=10	0.00	0.0000
Community_Area=11	0.00	0.0000
Community_Area=12	0.00	0.0000
Community_Area=13	0.00	0.0000
Community_Area=14	0.01	0.0000
Community_Area=15	0.01	0.0000
Community_Area=16	0.01	-0.0000
Community_Area=17	0.01	0.0000
Community_Area=18	0.00	0.0000
Community_Area=19	0.02	-0.0000
Community_Area=20	0.01	-0.0000
Community_Area=21	0.01	0.0000
Community_Area=22	0.02	-0.0000
Community_Area=23	0.03	0.0000
Community_Area=24	0.03	0.1667
Community_Area=25	0.06	0.0000
Community_Area=26	0.02	0.0000
Community_Area=27	0.02	0.0000
Community_Area=28	0.04	0.0000
Community_Area=29	0.03	0.0000
Community_Area=30	0.02	0.0000
Community_Area=31	0.01	0.0000
Community_Area=32	0.04	0.0000
Community_Area=33	0.01	0.0000
Community_Area=34	0.00	0.0000
Community_Area=35	0.01	0.0000
Community_Area=36	0.00	0.0000
Community_Area=37	0.00	-0.0000
Community_Area=38	0.01	0.0000
Community_Area=39	0.01	-0.0000
Community_Area=40	0.01	0.0000
Community_Area=41	0.01	0.0000
Community_Area=42	0.01	0.0000
Community_Area=43	0.03	0.0000
Community_Area=44	0.03	-0.0000
Community_Area=45	0.01	0.0000
Community_Area=46	0.02	0.0000
Community_Area=47	0.00	0.0000
Community_Area=48	0.01	-0.0000
Community_Area=49	0.03	0.0000

Variable	Segment 1	Segment 2
Community_Area=50	0.00	0.0000
Community_Area=51	0.01	0.0000
Community_Area=52	0.00	0.0000
Community_Area=53	0.02	0.0000
Community_Area=54	0.01	0.0000
Community_Area=55	0.00	0.0000
Community_Area=56	0.01	0.0000
Community_Area=57	0.00	-0.0000
Community_Area=58	0.01	-0.0000
Community_Area=59	0.00	0.0000
Community_Area=60	0.01	-0.0000
Community_Area=61	0.02	0.0000
Community_Area=62	0.00	0.0000
Community_Area=63	0.01	0.0000
Community_Area=64	0.00	0.0000
Community_Area=65	0.01	0.0000
Community_Area=66	0.02	0.0000
Community_Area=67	0.03	0.0000
Community_Area=68	0.02	0.0000
Community_Area=69	0.03	-0.0000
Community_Area=70	0.01	-0.0000
Community_Area=71	0.03	-0.0000
Community_Area=72	0.00	0.0000
Community_Area=73	0.01	0.0000
Community_Area=74	0.00	-0.0000
Community_Area=75	0.01	0.0000
Community_Area=76	0.01	0.6667
Community_Area=77	0.01	0.0000
District=1	0.05	0.0000
District=2	0.04	-0.0000
District=3	0.05	0.0000
District=4	0.05	-0.0000
District=5	0.05	-0.0000
District=6	0.06	-0.0000
District=7	0.05	0.0000
District=8	0.06	0.0000
District=9	0.04	0.0000
District=10	0.05	0.0000
District=11	0.07	0.0000
District=12	0.05	0.0000
District=14	0.03	0.0000
District=15	0.04	0.0000
District=16	0.03	0.0000
District=17	0.03	0.0000
District=18	0.05	0.0000
District=19	0.04	0.0000
District=20	0.02	0.0000
District=22	0.03	0.0000
District=24	0.03	0.0000
District=25	0.05	0.0000
District=31	0.00	1.0000

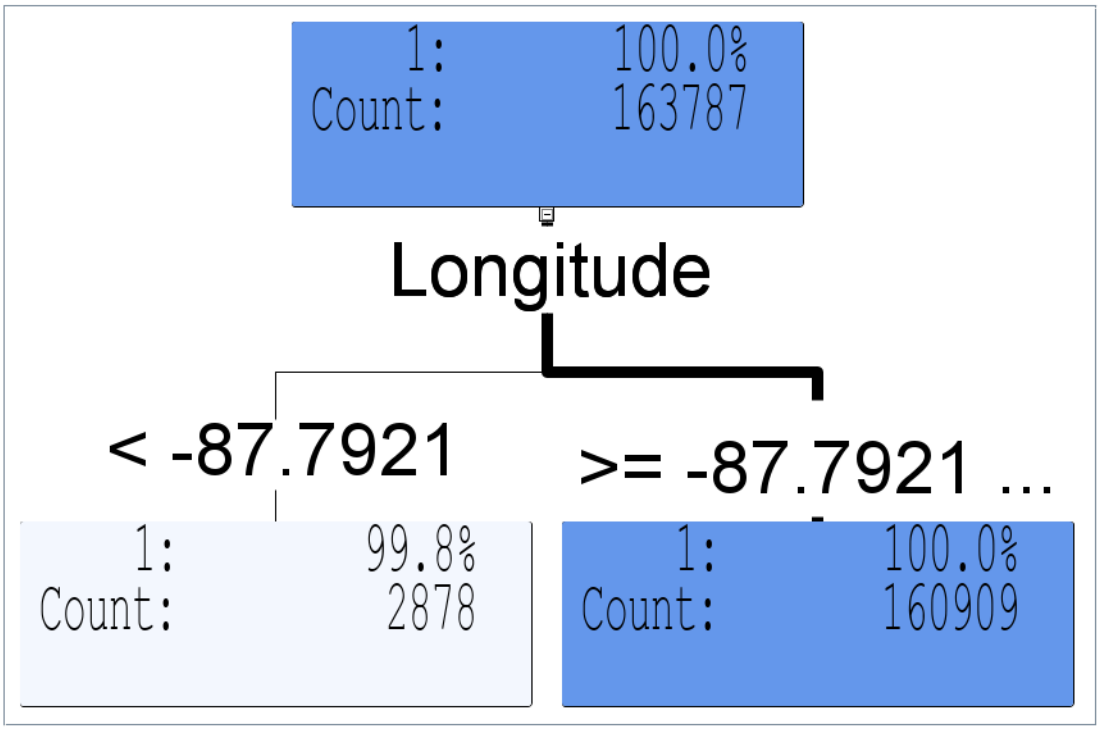
Variable	Segment 1	Segment 2
Domestic=0	0.80	1.0000
Domestic=1	0.20	0.0000
Ward=1	0.02	0.1667
Ward=2	0.03	0.0000
Ward=3	0.03	0.0000
Ward=4	0.02	0.0000
Ward=5	0.02	0.0000
Ward=6	0.04	0.0000
Ward=7	0.03	0.0000
Ward=8	0.03	0.0000
Ward=9	0.03	0.0000
Ward=10	0.01	0.0000
Ward=11	0.01	0.0000
Ward=12	0.01	0.0000
Ward=13	0.01	-0.0000
Ward=14	0.01	0.0000
Ward=15	0.02	0.0000
Ward=16	0.03	-0.0000
Ward=17	0.03	0.0000
Ward=18	0.01	0.0000
Ward=19	0.01	-0.0000
Ward=20	0.03	0.0000
Ward=21	0.03	0.0000
Ward=22	0.01	-0.0000
Ward=23	0.01	0.0000
Ward=24	0.04	0.0000
Ward=25	0.02	-0.0000
Ward=26	0.01	0.0000
Ward=27	0.05	-0.0000
Ward=28	0.05	-0.0000
Ward=29	0.02	0.0000
Ward=30	0.01	-0.0000
Ward=31	0.01	0.0000
Ward=32	0.01	0.0000
Ward=33	0.01	0.0000
Ward=34	0.03	0.0000
Ward=35	0.01	0.0000
Ward=36	0.01	0.0000
Ward=37	0.03	0.0000
Ward=38	0.01	0.0000
Ward=39	0.01	0.0000
Ward=40	0.01	0.0000
Ward=41	0.01	0.8333
Ward=42	0.06	0.0000
Ward=43	0.01	0.0000
Ward=44	0.01	0.0000
Ward=45	0.01	0.0000
Ward=46	0.01	0.0000
Ward=47	0.01	0.0000
Ward=48	0.01	0.0000

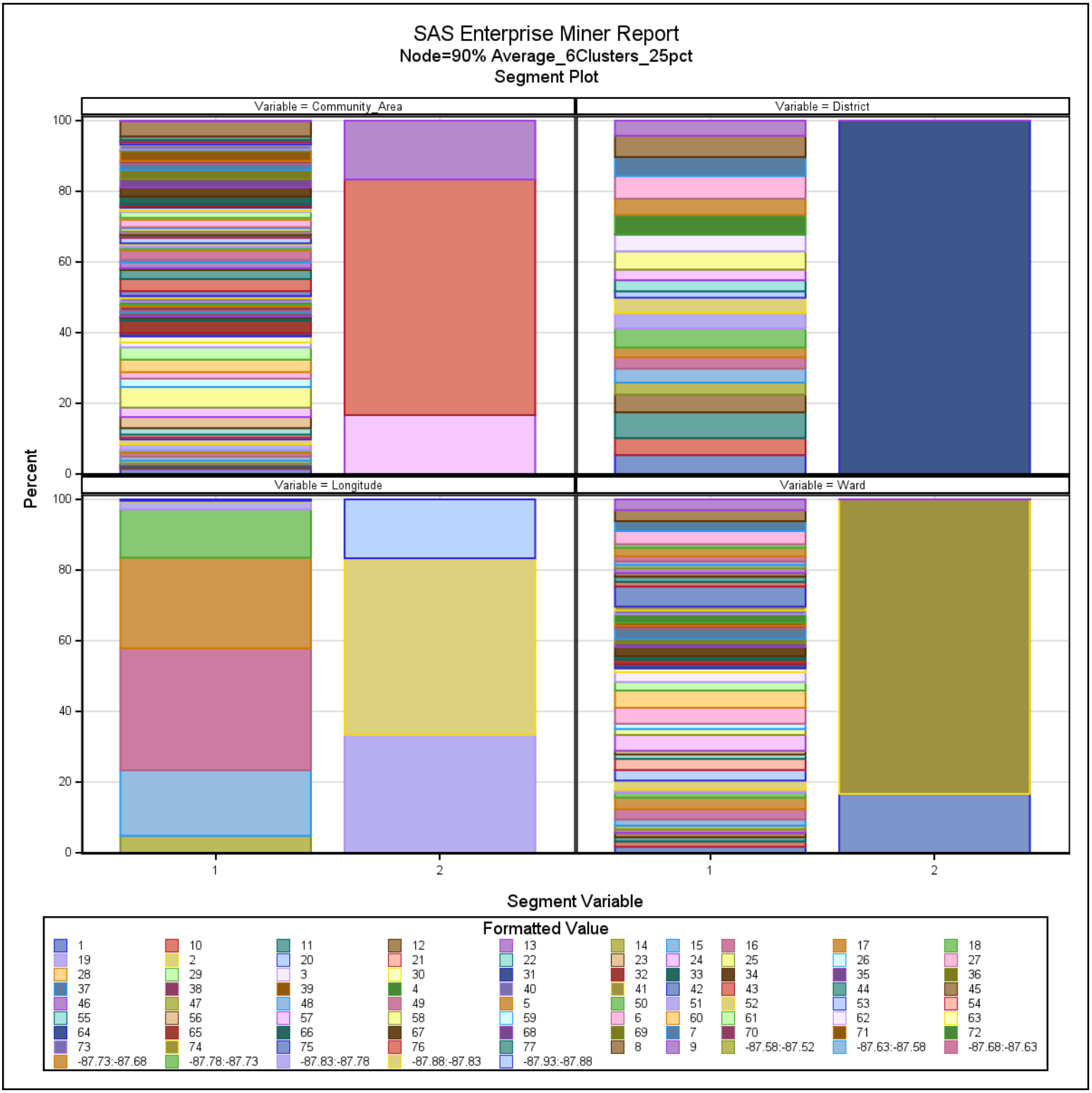
Variable	Segment 1	Segment 2
Ward=49	0.01	-0.0000
Ward=50	0.01	0.0000

SAS Enterprise Miner Report
Node=90% Average_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=90% Average_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=90% Average_6Clusters_25pct
Segment Plot
Variable = Community_Area
Variable = District
Variable = Longitude
Variable = Ward
Percent
0
20
40
60
80
100
1
2
Segment Variable
Formatted Value
1
10
11
12
13
14
15
16
17
18
19
2
20
21
22
23
24
25
26
27
28
29
3
30
31
32
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60
61
62
63
64
65
66
67
68
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7
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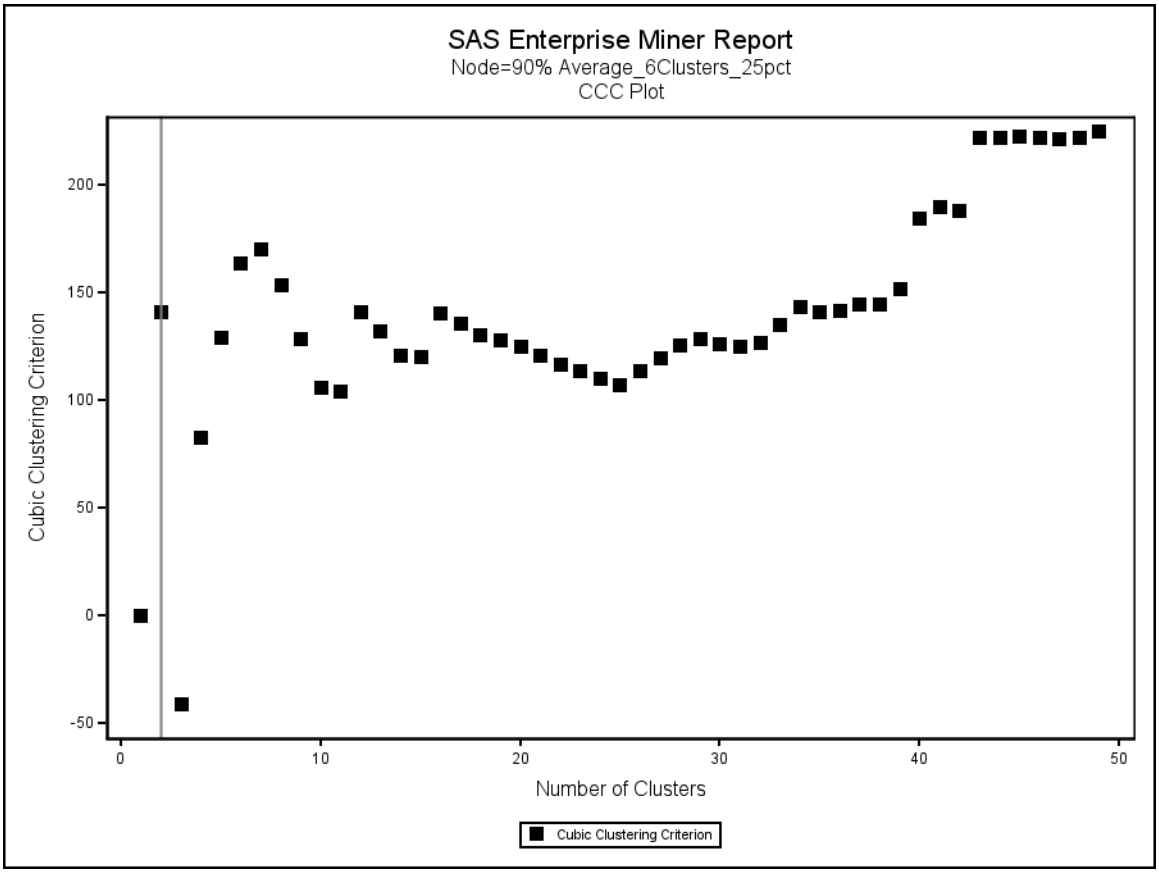
71
72
73
74
75
76
77
8
9
-87.58:-87.52
-87.63:-87.58
-87.68:-87.63
-87.73:-87.68
-87.78:-87.73
-87.83:-87.78
-87.88:-87.83
-87.93:-87.88


SAS Enterprise Miner Report
Node=90% Average_6Clusters_25pct
CCC Plot
Cubic Clustering Criterion
200
150
100
50
0
-50
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=90% Centroid_6Clusters_25pct Summary

Node id = Clus4
Node label = 90% Centroid_6Clusters_25pct
Meta path = Ids => Smpl => Smpl2 => Clus4
Notes =

Node=90% Centroid_6Clusters_25pct Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	CENTROID	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=90% Centroid_6Clusters_25pct Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=90% Centroid_6Clusters_25pct Mean Statistics

Variable	Segment 1	Segment 2
Clustering Criterion	0.24	0.2381
Maximum Relative Change in Cluster Seeds	0.00	0.0000
Improvement in Clustering Criterion	0.00	0.0000
Frequency of Cluster	163781.00	6.0000
Root-Mean-Square Standard Deviation	0.24	0.2088
Maximum Distance from Cluster Seed	6.03	3.3346
Nearest Cluster	2.00	1.0000
Distance to Nearest Cluster	34.69	34.6917
Hour	13.01	9.8333
Latitude	41.84	41.9809
Longitude	-87.67	-87.8618
Month	6.54	8.0000

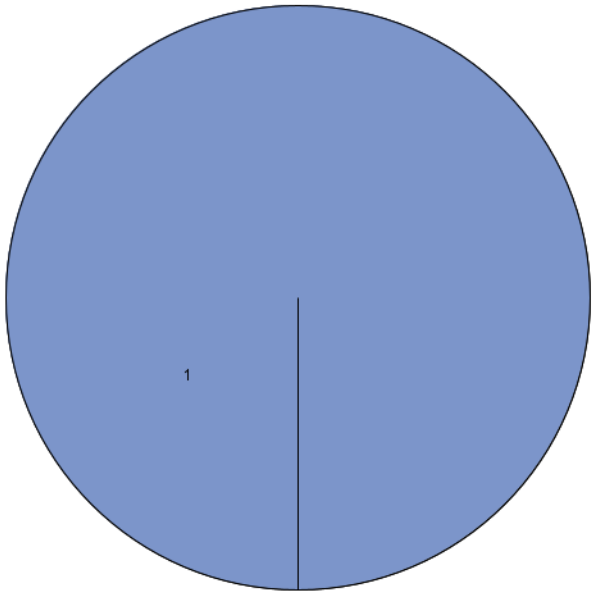
Variable	Segment 1	Segment 2
Arrest=0	0.80	0.8333
Arrest=1	0.20	0.1667
Community_Area=1	0.01	0.0000
Community_Area=2	0.01	0.0000
Community_Area=3	0.01	0.0000
Community_Area=4	0.01	0.0000
Community_Area=5	0.00	0.0000
Community_Area=6	0.02	0.0000
Community_Area=7	0.02	0.0000
Community_Area=8	0.04	-0.0000
Community_Area=9	0.00	0.1667
Community_Area=10	0.00	0.0000
Community_Area=11	0.00	0.0000
Community_Area=12	0.00	0.0000
Community_Area=13	0.00	0.0000
Community_Area=14	0.01	0.0000
Community_Area=15	0.01	0.0000
Community_Area=16	0.01	-0.0000
Community_Area=17	0.01	0.0000
Community_Area=18	0.00	0.0000
Community_Area=19	0.02	-0.0000
Community_Area=20	0.01	-0.0000
Community_Area=21	0.01	0.0000
Community_Area=22	0.02	-0.0000
Community_Area=23	0.03	0.0000
Community_Area=24	0.03	0.1667
Community_Area=25	0.06	0.0000
Community_Area=26	0.02	0.0000
Community_Area=27	0.02	0.0000
Community_Area=28	0.04	0.0000
Community_Area=29	0.03	0.0000
Community_Area=30	0.02	0.0000
Community_Area=31	0.01	0.0000
Community_Area=32	0.04	0.0000
Community_Area=33	0.01	0.0000
Community_Area=34	0.00	0.0000
Community_Area=35	0.01	0.0000
Community_Area=36	0.00	0.0000
Community_Area=37	0.00	-0.0000
Community_Area=38	0.01	0.0000
Community_Area=39	0.01	-0.0000
Community_Area=40	0.01	0.0000
Community_Area=41	0.01	0.0000
Community_Area=42	0.01	0.0000
Community_Area=43	0.03	0.0000
Community_Area=44	0.03	-0.0000
Community_Area=45	0.01	0.0000
Community_Area=46	0.02	0.0000
Community_Area=47	0.00	0.0000
Community_Area=48	0.01	-0.0000
Community_Area=49	0.03	0.0000

Variable	Segment 1	Segment 2
Community_Area=50	0.00	0.0000
Community_Area=51	0.01	0.0000
Community_Area=52	0.00	0.0000
Community_Area=53	0.02	0.0000
Community_Area=54	0.01	0.0000
Community_Area=55	0.00	0.0000
Community_Area=56	0.01	0.0000
Community_Area=57	0.00	-0.0000
Community_Area=58	0.01	-0.0000
Community_Area=59	0.00	0.0000
Community_Area=60	0.01	-0.0000
Community_Area=61	0.02	0.0000
Community_Area=62	0.00	0.0000
Community_Area=63	0.01	0.0000
Community_Area=64	0.00	0.0000
Community_Area=65	0.01	0.0000
Community_Area=66	0.02	0.0000
Community_Area=67	0.03	0.0000
Community_Area=68	0.02	0.0000
Community_Area=69	0.03	-0.0000
Community_Area=70	0.01	-0.0000
Community_Area=71	0.03	-0.0000
Community_Area=72	0.00	0.0000
Community_Area=73	0.01	0.0000
Community_Area=74	0.00	-0.0000
Community_Area=75	0.01	0.0000
Community_Area=76	0.01	0.6667
Community_Area=77	0.01	0.0000
District=1	0.05	0.0000
District=2	0.04	-0.0000
District=3	0.05	0.0000
District=4	0.05	-0.0000
District=5	0.05	-0.0000
District=6	0.06	-0.0000
District=7	0.05	0.0000
District=8	0.06	0.0000
District=9	0.04	0.0000
District=10	0.05	0.0000
District=11	0.07	0.0000
District=12	0.05	0.0000
District=14	0.03	0.0000
District=15	0.04	0.0000
District=16	0.03	0.0000
District=17	0.03	0.0000
District=18	0.05	0.0000
District=19	0.04	0.0000
District=20	0.02	0.0000
District=22	0.03	0.0000
District=24	0.03	0.0000
District=25	0.05	0.0000
District=31	0.00	1.0000

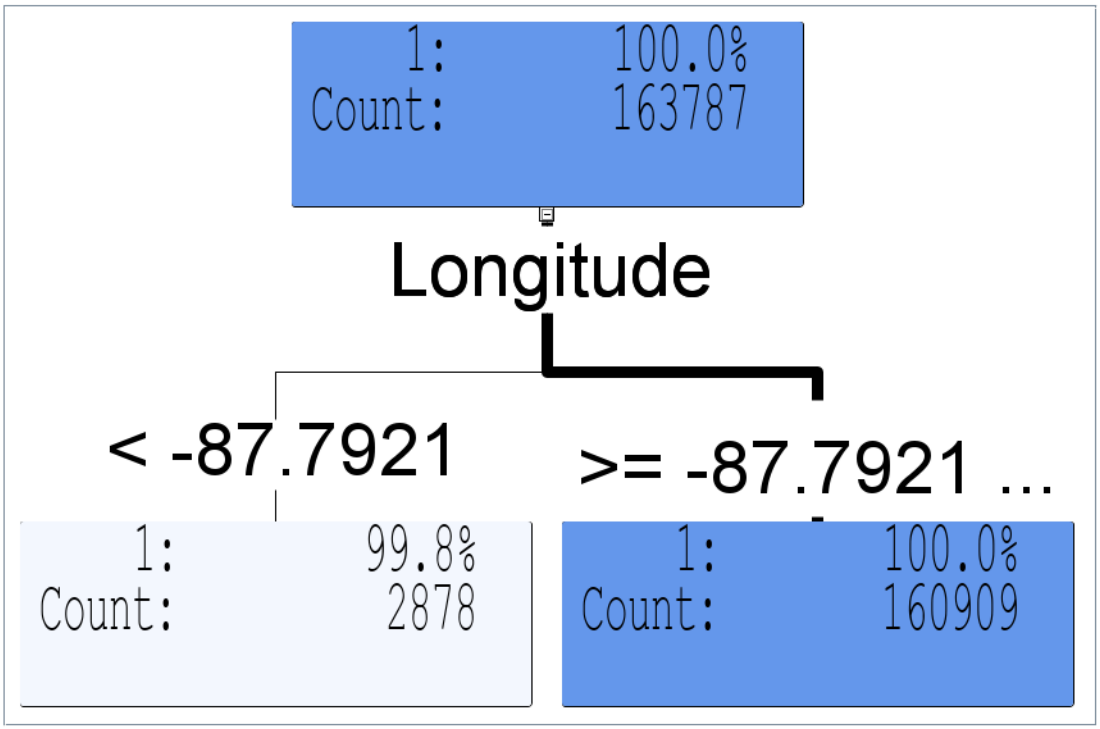
Variable	Segment 1	Segment 2
Domestic=0	0.80	1.0000
Domestic=1	0.20	0.0000
Ward=1	0.02	0.1667
Ward=2	0.03	0.0000
Ward=3	0.03	0.0000
Ward=4	0.02	0.0000
Ward=5	0.02	0.0000
Ward=6	0.04	0.0000
Ward=7	0.03	0.0000
Ward=8	0.03	0.0000
Ward=9	0.03	0.0000
Ward=10	0.01	0.0000
Ward=11	0.01	0.0000
Ward=12	0.01	0.0000
Ward=13	0.01	-0.0000
Ward=14	0.01	0.0000
Ward=15	0.02	0.0000
Ward=16	0.03	-0.0000
Ward=17	0.03	0.0000
Ward=18	0.01	0.0000
Ward=19	0.01	-0.0000
Ward=20	0.03	0.0000
Ward=21	0.03	0.0000
Ward=22	0.01	-0.0000
Ward=23	0.01	0.0000
Ward=24	0.04	0.0000
Ward=25	0.02	-0.0000
Ward=26	0.01	0.0000
Ward=27	0.05	-0.0000
Ward=28	0.05	-0.0000
Ward=29	0.02	0.0000
Ward=30	0.01	-0.0000
Ward=31	0.01	0.0000
Ward=32	0.01	0.0000
Ward=33	0.01	0.0000
Ward=34	0.03	0.0000
Ward=35	0.01	0.0000
Ward=36	0.01	0.0000
Ward=37	0.03	0.0000
Ward=38	0.01	0.0000
Ward=39	0.01	0.0000
Ward=40	0.01	0.0000
Ward=41	0.01	0.8333
Ward=42	0.06	0.0000
Ward=43	0.01	0.0000
Ward=44	0.01	0.0000
Ward=45	0.01	0.0000
Ward=46	0.01	0.0000
Ward=47	0.01	0.0000
Ward=48	0.01	0.0000

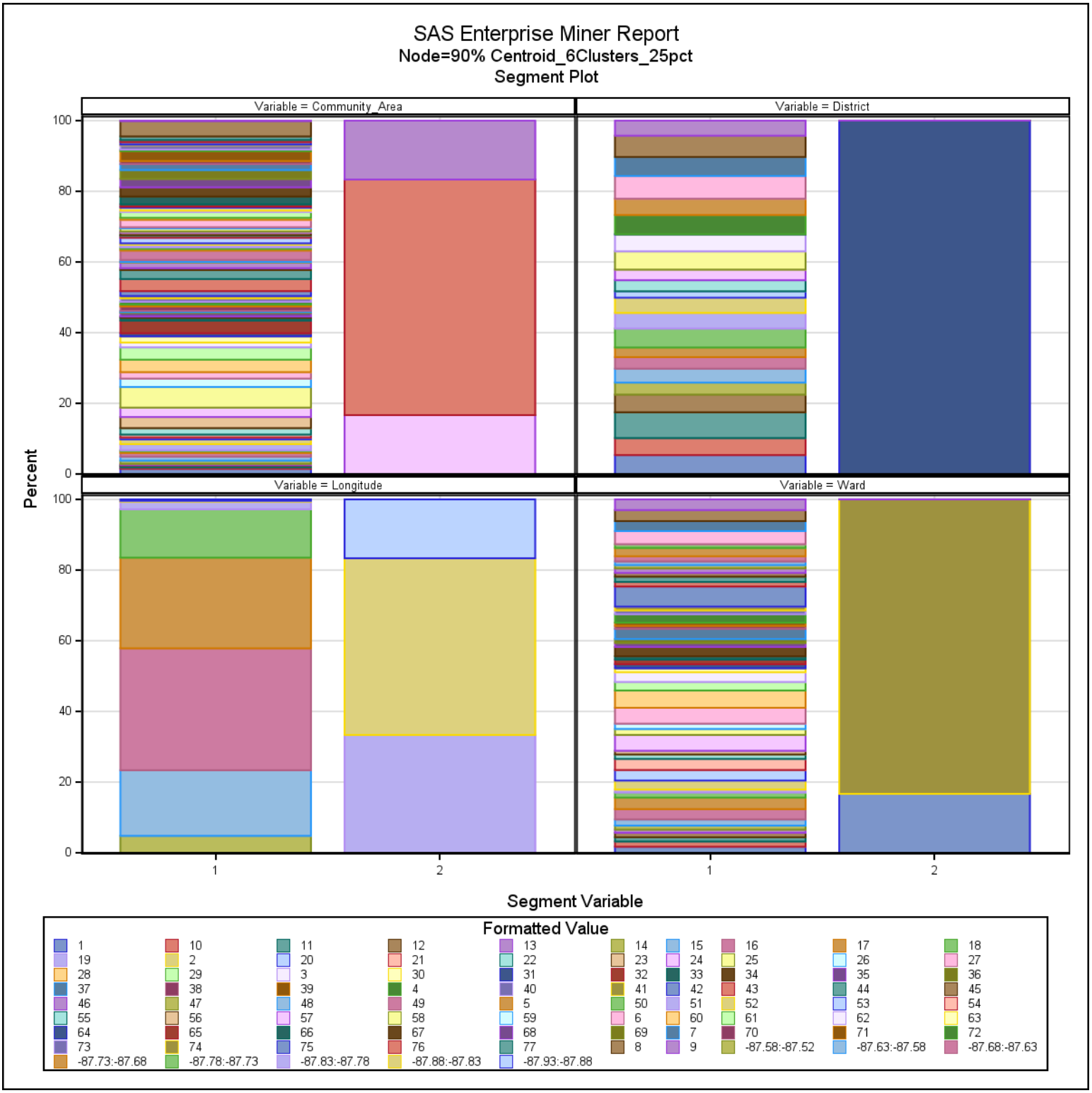
Variable	Segment 1	Segment 2
Ward=49	0.01	-0.0000
Ward=50	0.01	0.0000

SAS Enterprise Miner Report
Node=90% Centroid_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=90% Centroid_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=90% Centroid_6Clusters_25pct
Segment Plot
Variable = Community_Area
Variable = District
Variable = Longitude
Variable = Ward
Percent
0
20
40
60
80
100
1
2
Segment Variable
Formatted Value
1
10
11
12
13
14
15
16
17
18
19
2
20
21
22
23
24
25
26
27
28
29
3
30
31
32
33
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37
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4
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47
48
49
5
50
51
52
53
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58
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6
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61
62
63
64
65
66
67
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72
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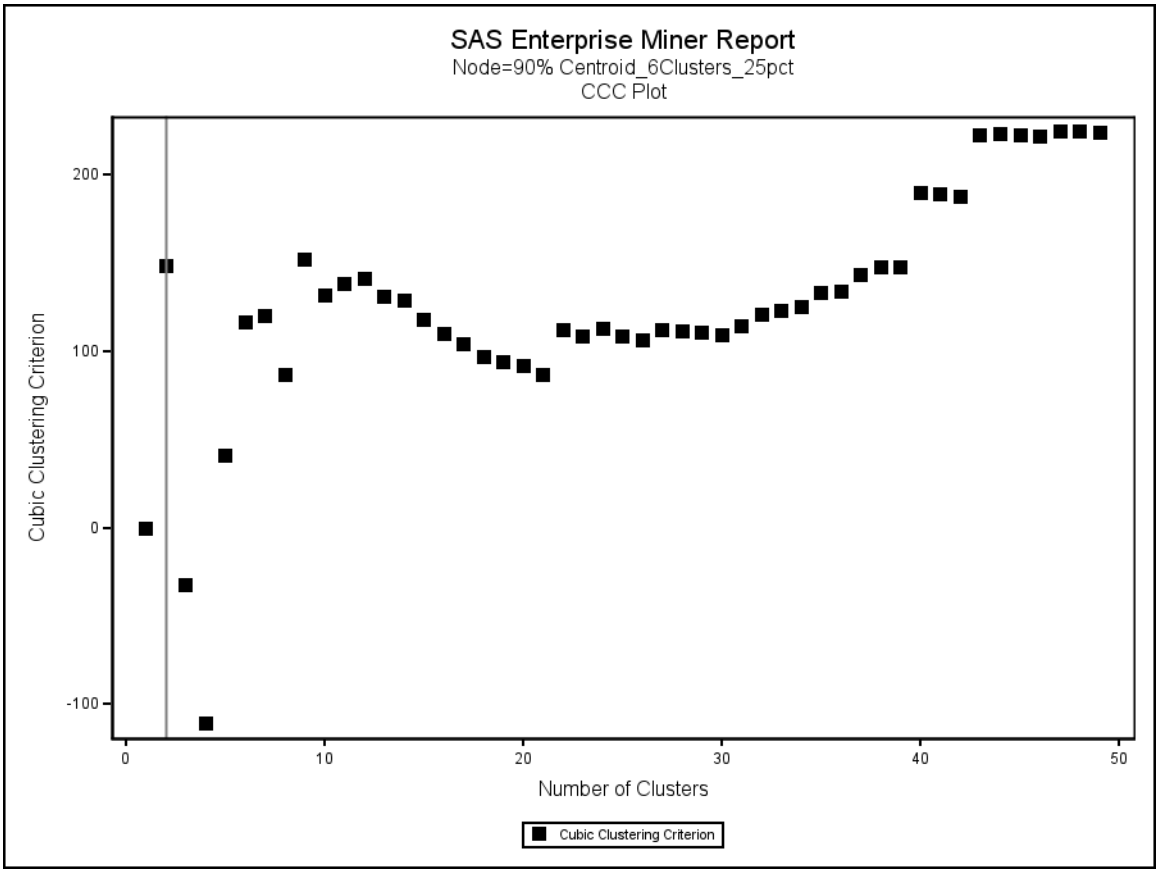
74
75
76
77
8
9
-87.58:-87.52
-87.63:-87.58
-87.68:-87.63
-87.73:-87.68
-87.78:-87.73
-87.83:-87.78
-87.88:-87.83
-87.93:-87.88


SAS Enterprise Miner Report
Node=90% Centroid_6Clusters_25pct
CCC Plot
Cubic Clustering Criterion
200
100
0
-100
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=90% Ward_6Clusters_25pct Summary

Node id = Clus3
Node label = 90% Ward_6Clusters_25pct
Meta path = Ids => Smpl => Smpl2 => Clus3
Notes =

Node=90% Ward_6Clusters_25pct Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	WARD		ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=90% Ward_6Clusters_25pct Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=90% Ward_6Clusters_25pct Mean Statistics

Variable	Segment 1	Segment 2	Segment 3
Clustering Criterion	0.2219	0.22	0.22
Maximum Relative Change in Cluster Seeds	0.0091	0.01	0.01
Improvement in Clustering Criterion	.	.	.
Frequency of Cluster	6.0000	69205.00	94576.00
Root-Mean-Square Standard Deviation	0.2088	0.22	0.22
Maximum Distance from Cluster Seed	3.3346	5.14	5.70
Nearest Cluster	3.0000	3.00	2.00
Distance to Nearest Cluster	34.6186	2.20	2.20
Hour	9.8333	12.92	13.07
Latitude	41.9809	41.76	41.90
Longitude	-87.8618	-87.63	-87.70
Month	8.0000	6.54	6.55

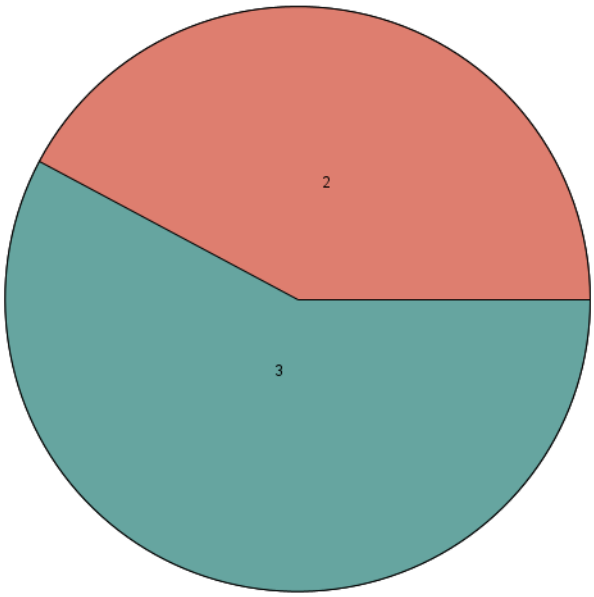
Variable	Segment 1	Segment 2	Segment 3
Arrest=0	0.8333	0.80	0.81
Arrest=1	0.1667	0.20	0.19
Community_Area=1	0.0000	-0.00	0.03
Community_Area=2	0.0000	-0.00	0.02
Community_Area=3	0.0000	0.00	0.02
Community_Area=4	0.0000	-0.00	0.01
Community_Area=5	0.0000	-0.00	0.01
Community_Area=6	0.0000	0.00	0.04
Community_Area=7	0.0000	-0.00	0.03
Community_Area=8	-0.0000	0.00	0.07
Community_Area=9	0.1667	-0.00	0.00
Community_Area=10	0.0000	0.00	0.01
Community_Area=11	0.0000	0.00	0.01
Community_Area=12	0.0000	-0.00	0.00
Community_Area=13	0.0000	0.00	0.01
Community_Area=14	0.0000	-0.00	0.02
Community_Area=15	0.0000	0.00	0.02
Community_Area=16	-0.0000	-0.00	0.02
Community_Area=17	0.0000	-0.00	0.01
Community_Area=18	0.0000	0.00	0.00
Community_Area=19	-0.0000	0.00	0.03
Community_Area=20	-0.0000	0.00	0.01
Community_Area=21	0.0000	-0.00	0.01
Community_Area=22	-0.0000	0.00	0.03
Community_Area=23	0.0000	0.00	0.05
Community_Area=24	0.1667	0.00	0.05
Community_Area=25	0.0000	0.00	0.10
Community_Area=26	0.0000	0.00	0.04
Community_Area=27	0.0000	0.00	0.03
Community_Area=28	0.0000	0.00	0.06
Community_Area=29	0.0000	-0.00	0.06
Community_Area=30	0.0000	0.00	0.03
Community_Area=31	0.0000	0.00	0.01
Community_Area=32	0.0000	0.00	0.06
Community_Area=33	0.0000	0.02	0.00
Community_Area=34	0.0000	0.01	-0.00
Community_Area=35	0.0000	0.02	0.00
Community_Area=36	0.0000	0.01	0.00
Community_Area=37	-0.0000	0.01	0.00
Community_Area=38	0.0000	0.03	-0.00
Community_Area=39	-0.0000	0.01	-0.00
Community_Area=40	0.0000	0.02	-0.00
Community_Area=41	0.0000	0.02	0.00
Community_Area=42	0.0000	0.03	0.00
Community_Area=43	0.0000	0.08	0.00
Community_Area=44	-0.0000	0.06	0.00
Community_Area=45	0.0000	0.01	-0.00
Community_Area=46	0.0000	0.04	-0.00
Community_Area=47	0.0000	0.00	0.00
Community_Area=48	-0.0000	0.01	0.00
Community_Area=49	0.0000	0.06	-0.00

Variable	Segment 1	Segment 2	Segment 3
Community_Area=50	0.0000	0.01	0.00
Community_Area=51	0.0000	0.02	0.00
Community_Area=52	0.0000	0.01	0.00
Community_Area=53	0.0000	0.04	-0.00
Community_Area=54	0.0000	0.01	0.00
Community_Area=55	0.0000	0.01	-0.00
Community_Area=56	0.0000	-0.00	0.01
Community_Area=57	-0.0000	0.00	0.01
Community_Area=58	-0.0000	0.00	0.01
Community_Area=59	0.0000	0.00	0.00
Community_Area=60	-0.0000	0.01	0.00
Community_Area=61	0.0000	0.04	0.00
Community_Area=62	0.0000	0.00	0.01
Community_Area=63	0.0000	0.01	0.01
Community_Area=64	0.0000	0.00	0.01
Community_Area=65	0.0000	0.00	0.01
Community_Area=66	0.0000	0.05	0.00
Community_Area=67	0.0000	0.06	0.00
Community_Area=68	0.0000	0.06	-0.00
Community_Area=69	-0.0000	0.06	0.00
Community_Area=70	-0.0000	0.02	0.00
Community_Area=71	-0.0000	0.07	-0.00
Community_Area=72	0.0000	0.01	-0.00
Community_Area=73	0.0000	0.03	-0.00
Community_Area=74	-0.0000	0.00	-0.00
Community_Area=75	0.0000	0.02	0.00
Community_Area=76	0.6667	0.00	0.01
Community_Area=77	0.0000	0.00	0.02
District=1	0.0000	0.03	0.07
District=2	-0.0000	0.10	-0.00
District=3	0.0000	0.11	0.00
District=4	-0.0000	0.13	0.00
District=5	-0.0000	0.11	-0.00
District=6	-0.0000	0.15	-0.00
District=7	0.0000	0.13	-0.00
District=8	0.0000	0.08	0.04
District=9	0.0000	0.07	0.02
District=10	0.0000	0.00	0.08
District=11	0.0000	-0.00	0.13
District=12	0.0000	0.00	0.08
District=14	0.0000	-0.00	0.06
District=15	0.0000	-0.00	0.07
District=16	0.0000	0.00	0.06
District=17	0.0000	0.00	0.05
District=18	0.0000	0.00	0.09
District=19	0.0000	0.00	0.08
District=20	0.0000	-0.00	0.03
District=22	0.0000	0.07	-0.00
District=24	0.0000	0.00	0.05
District=25	0.0000	-0.00	0.09
District=31	1.0000	-0.00	-0.00

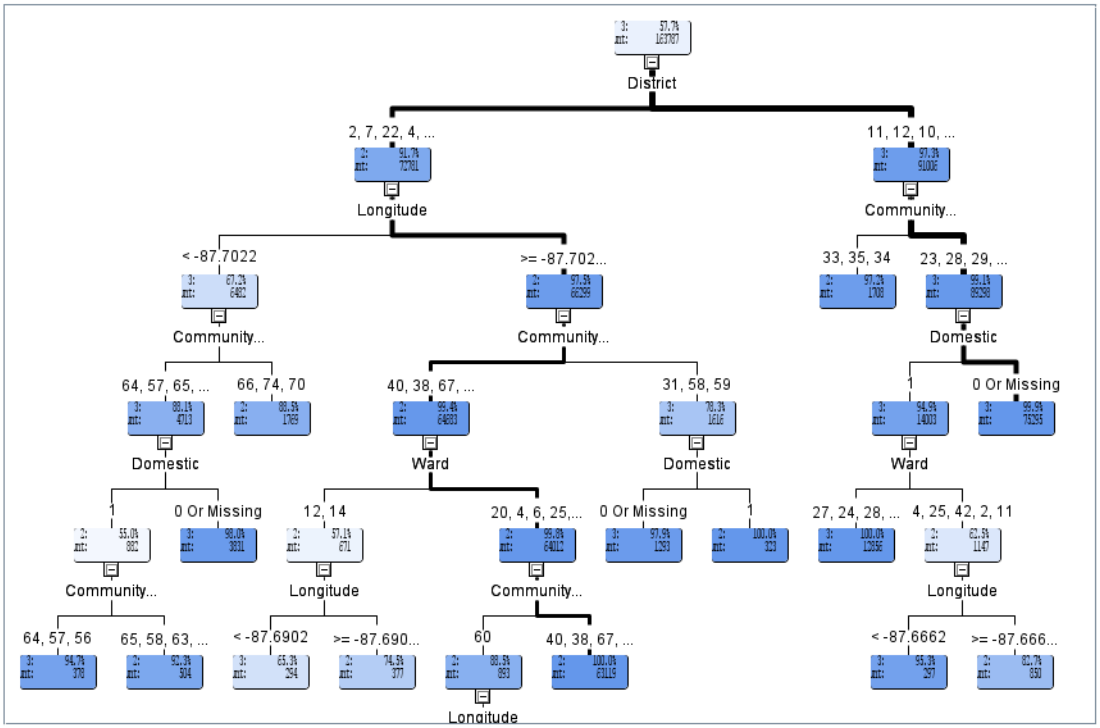
Variable	Segment 1	Segment 2	Segment 3
Domestic=0	1.0000	0.72	0.86
Domestic=1	0.0000	0.28	0.14
Ward=1	0.1667	-0.00	0.03
Ward=2	0.0000	0.01	0.04
Ward=3	0.0000	0.07	0.00
Ward=4	0.0000	0.05	0.01
Ward=5	0.0000	0.06	0.00
Ward=6	0.0000	0.09	-0.00
Ward=7	0.0000	0.07	-0.00
Ward=8	0.0000	0.07	-0.00
Ward=9	0.0000	0.07	-0.00
Ward=10	0.0000	0.04	-0.00
Ward=11	0.0000	0.02	0.01
Ward=12	0.0000	0.00	0.02
Ward=13	-0.0000	0.00	0.01
Ward=14	0.0000	0.01	0.02
Ward=15	0.0000	0.04	0.00
Ward=16	-0.0000	0.07	0.00
Ward=17	0.0000	0.08	-0.00
Ward=18	0.0000	0.03	0.00
Ward=19	-0.0000	0.02	-0.00
Ward=20	0.0000	0.07	-0.00
Ward=21	0.0000	0.07	0.00
Ward=22	-0.0000	0.00	0.02
Ward=23	0.0000	0.00	0.02
Ward=24	0.0000	-0.00	0.08
Ward=25	-0.0000	0.01	0.02
Ward=26	0.0000	0.00	0.03
Ward=27	-0.0000	0.00	0.08
Ward=28	-0.0000	0.00	0.08
Ward=29	0.0000	-0.00	0.04
Ward=30	-0.0000	0.00	0.02
Ward=31	0.0000	-0.00	0.02
Ward=32	0.0000	-0.00	0.02
Ward=33	0.0000	-0.00	0.02
Ward=34	0.0000	0.06	0.00
Ward=35	0.0000	0.00	0.02
Ward=36	0.0000	-0.00	0.02
Ward=37	0.0000	0.00	0.05
Ward=38	0.0000	-0.00	0.01
Ward=39	0.0000	-0.00	0.02
Ward=40	0.0000	-0.00	0.02
Ward=41	0.8333	-0.00	0.02
Ward=42	0.0000	0.00	0.10
Ward=43	0.0000	-0.00	0.02
Ward=44	0.0000	0.00	0.03
Ward=45	0.0000	-0.00	0.02
Ward=46	0.0000	-0.00	0.02
Ward=47	0.0000	0.00	0.02
Ward=48	0.0000	-0.00	0.02

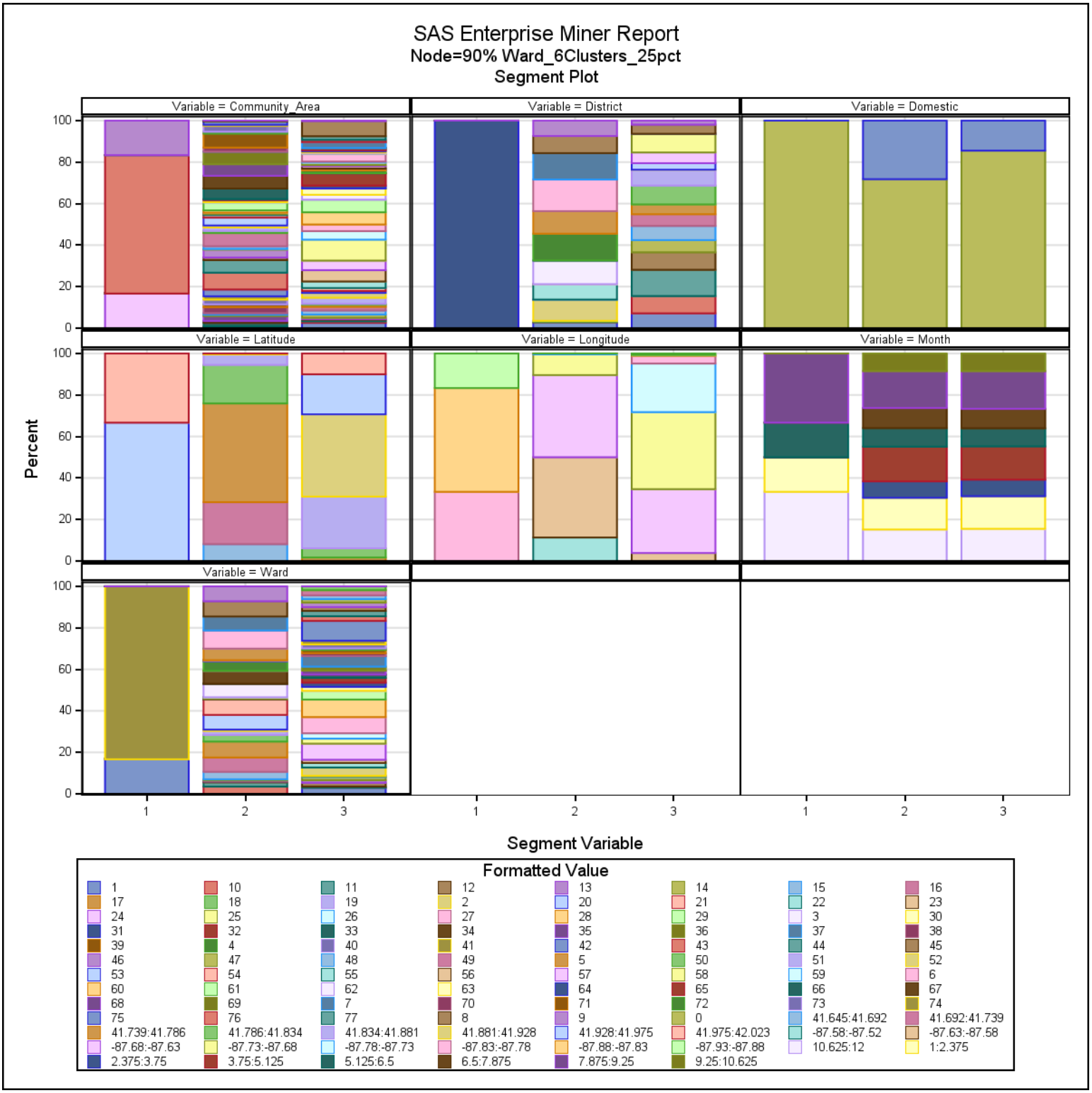
Variable	Segment 1	Segment 2	Segment 3
Ward=49	-0.0000	-0.00	0.03
Ward=50	0.0000	-0.00	0.02

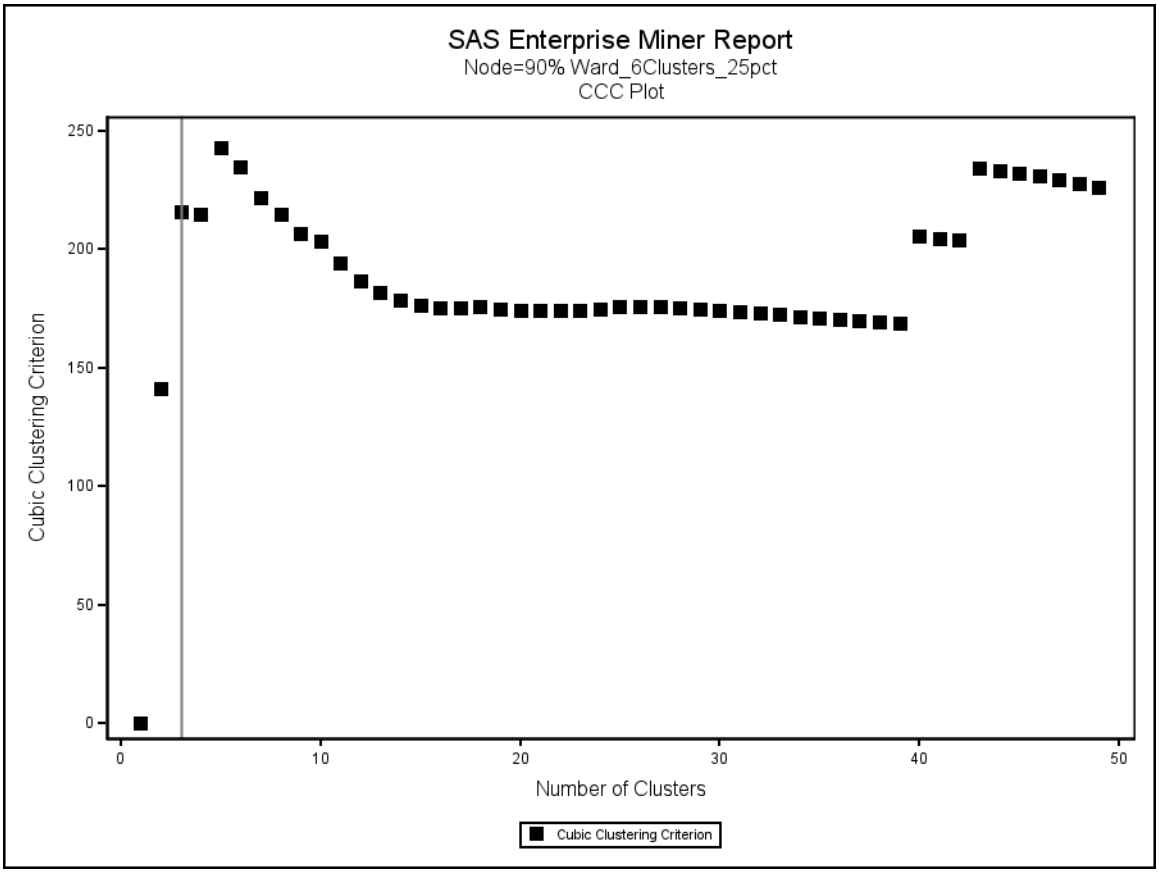
SAS Enterprise Miner Report
Node=90% Ward_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=90% Ward_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=90% Ward_6Clusters_25pct
Segment Plot
Variable = Community_Area
Variable = District
Variable = Domestic
Variable = Latitude
Variable = Longitude
Variable = Month
Variable = Ward
Percent
Segment Variable
Formatted Value
1
10
11
12
13
14
15
16
17
18
19
2
20
21
22
23
24
25
26
27
28
29
3
30
31
32
33
34
35
36
37
38
39
4
40
41
42
43
44
45
46
47
48
49
5
50
51
52
53
54
55
56
57
58
59
6
60
61
62
63
64
65
66
67
68
69
7
70
71
72
73
74
75
76
77
8
9
0
41.645:41.692
41.692:41.739
41.739:41.786
41.786:41.834
41.834:41.881
41.881:41.928
41.928:41.975
41.975:42.023
-87.58:-87.52
-87.63:-87.58
-87.68:-87.63
-87.73:-87.68
-87.78:-87.73
-87.83:-87.78
-87.88:-87.83
-87.93:-87.88
10.625:12
1:2.375
2.375:3.75
3.75:5.125
5.125:6.5
6.5:7.875
7.875:9.25
9.25:10.625


SAS Enterprise Miner Report
Node=90% Ward_6Clusters_25pct
CCC Plot
Cubic Clustering Criterion
250
200
150
100
50
0
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=Centroid_6Clusters_25pct
Summary

Node id = Clus2
Node label = Centroid_6Clusters_25pct
Meta path = Ids => Smpl => Clus2
Notes =

Node=Centroid_6Clusters_25pct
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	CENTROID	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=Centroid_6Clusters_25pct
Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=Centroid_6Clusters_25pct
Mean Statistics

Variable	Segment 1	Segment 2
Clustering Criterion	0.24	0.2381
Maximum Relative Change in Cluster Seeds	0.00	0.0000
Improvement in Clustering Criterion	0.00	0.0000
Frequency of Cluster	181980.00	6.0000
Root-Mean-Square Standard Deviation	0.24	0.2091
Maximum Distance from Cluster Seed	6.04	3.3382
Nearest Cluster	2.00	1.0000
Distance to Nearest Cluster	36.54	36.5434
Hour	13.01	9.8333
Latitude	41.84	41.9809
Longitude	-87.67	-87.8618
Month	6.55	8.0000

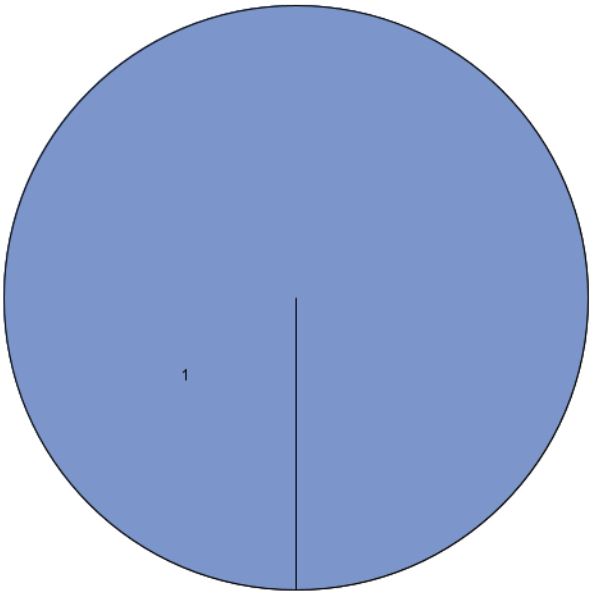
Variable	Segment 1	Segment 2
Arrest=0	0.80	0.8333
Arrest=1	0.20	0.1667
Community_Area=1	0.01	0.0000
Community_Area=2	0.01	0.0000
Community_Area=3	0.01	-0.0000
Community_Area=4	0.01	0.0000
Community_Area=5	0.00	-0.0000
Community_Area=6	0.02	0.0000
Community_Area=7	0.02	0.0000
Community_Area=8	0.04	-0.0000
Community_Area=9	0.00	0.1667
Community_Area=10	0.00	-0.0000
Community_Area=11	0.00	-0.0000
Community_Area=12	0.00	0.0000
Community_Area=13	0.00	0.0000
Community_Area=14	0.01	0.0000
Community_Area=15	0.01	0.0000
Community_Area=16	0.01	0.0000
Community_Area=17	0.01	0.0000
Community_Area=18	0.00	-0.0000
Community_Area=19	0.02	-0.0000
Community_Area=20	0.01	0.0000
Community_Area=21	0.01	0.0000
Community_Area=22	0.02	0.0000
Community_Area=23	0.03	-0.0000
Community_Area=24	0.03	0.1667
Community_Area=25	0.06	0.0000
Community_Area=26	0.02	0.0000
Community_Area=27	0.02	0.0000
Community_Area=28	0.04	0.0000
Community_Area=29	0.04	0.0000
Community_Area=30	0.02	0.0000
Community_Area=31	0.01	0.0000
Community_Area=32	0.04	0.0000
Community_Area=33	0.01	0.0000
Community_Area=34	0.00	0.0000
Community_Area=35	0.01	-0.0000
Community_Area=36	0.00	0.0000
Community_Area=37	0.00	0.0000
Community_Area=38	0.01	0.0000
Community_Area=39	0.01	0.0000
Community_Area=40	0.01	-0.0000
Community_Area=41	0.01	0.0000
Community_Area=42	0.01	-0.0000
Community_Area=43	0.03	0.0000
Community_Area=44	0.03	-0.0000
Community_Area=45	0.01	0.0000
Community_Area=46	0.02	0.0000
Community_Area=47	0.00	-0.0000
Community_Area=48	0.01	0.0000
Community_Area=49	0.03	0.0000

Variable	Segment 1	Segment 2
Community_Area=50	0.00	0.0000
Community_Area=51	0.01	0.0000
Community_Area=52	0.00	0.0000
Community_Area=53	0.02	0.0000
Community_Area=54	0.01	0.0000
Community_Area=55	0.00	0.0000
Community_Area=56	0.01	0.0000
Community_Area=57	0.00	-0.0000
Community_Area=58	0.01	0.0000
Community_Area=59	0.00	0.0000
Community_Area=60	0.01	0.0000
Community_Area=61	0.02	0.0000
Community_Area=62	0.00	0.0000
Community_Area=63	0.01	0.0000
Community_Area=64	0.00	0.0000
Community_Area=65	0.01	0.0000
Community_Area=66	0.02	0.0000
Community_Area=67	0.03	0.0000
Community_Area=68	0.02	0.0000
Community_Area=69	0.03	0.0000
Community_Area=70	0.01	0.0000
Community_Area=71	0.03	-0.0000
Community_Area=72	0.00	0.0000
Community_Area=73	0.01	0.0000
Community_Area=74	0.00	0.0000
Community_Area=75	0.01	-0.0000
Community_Area=76	0.01	0.6667
Community_Area=77	0.01	-0.0000
District=1	0.05	0.0000
District=2	0.04	0.0000
District=3	0.05	-0.0000
District=4	0.05	0.0000
District=5	0.05	0.0000
District=6	0.06	0.0000
District=7	0.05	0.0000
District=8	0.06	-0.0000
District=9	0.04	-0.0000
District=10	0.05	0.0000
District=11	0.07	0.0000
District=12	0.05	0.0000
District=14	0.03	0.0000
District=15	0.04	0.0000
District=16	0.03	0.0000
District=17	0.03	0.0000
District=18	0.05	-0.0000
District=19	0.04	0.0000
District=20	0.02	0.0000
District=22	0.03	0.0000
District=24	0.03	-0.0000
District=25	0.05	0.0000
District=31	-0.00	1.0000

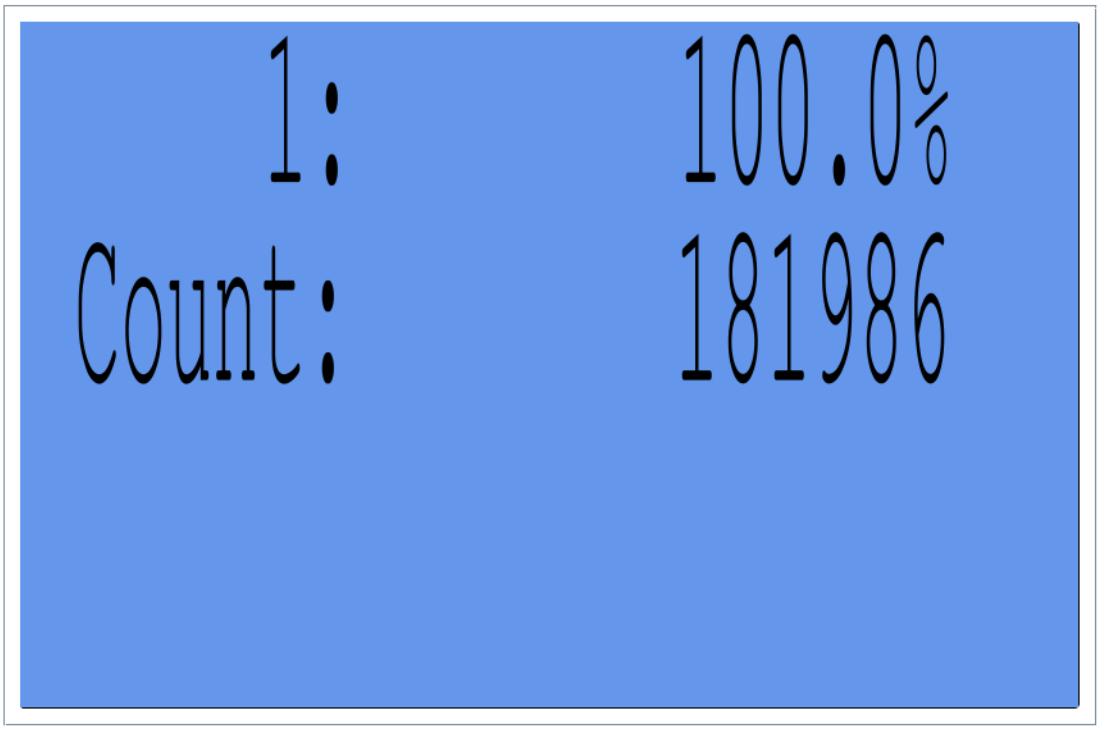
Variable	Segment 1	Segment 2
Domestic=0	0.80	1.0000
Domestic=1	0.20	-0.0000
Ward=1	0.02	0.1667
Ward=2	0.03	0.0000
Ward=3	0.03	0.0000
Ward=4	0.02	0.0000
Ward=5	0.02	-0.0000
Ward=6	0.04	0.0000
Ward=7	0.03	0.0000
Ward=8	0.03	0.0000
Ward=9	0.03	0.0000
Ward=10	0.01	0.0000
Ward=11	0.01	0.0000
Ward=12	0.01	-0.0000
Ward=13	0.01	0.0000
Ward=14	0.01	0.0000
Ward=15	0.02	0.0000
Ward=16	0.03	-0.0000
Ward=17	0.03	0.0000
Ward=18	0.01	-0.0000
Ward=19	0.01	0.0000
Ward=20	0.03	0.0000
Ward=21	0.03	0.0000
Ward=22	0.01	-0.0000
Ward=23	0.01	0.0000
Ward=24	0.04	0.0000
Ward=25	0.02	-0.0000
Ward=26	0.02	-0.0000
Ward=27	0.04	0.0000
Ward=28	0.05	0.0000
Ward=29	0.02	0.0000
Ward=30	0.01	0.0000
Ward=31	0.01	-0.0000
Ward=32	0.01	0.0000
Ward=33	0.01	0.0000
Ward=34	0.03	0.0000
Ward=35	0.01	0.0000
Ward=36	0.01	0.0000
Ward=37	0.03	0.0000
Ward=38	0.01	0.0000
Ward=39	0.01	0.0000
Ward=40	0.01	0.0000
Ward=41	0.01	0.8333
Ward=42	0.06	-0.0000
Ward=43	0.01	0.0000
Ward=44	0.01	0.0000
Ward=45	0.01	-0.0000
Ward=46	0.01	0.0000
Ward=47	0.01	0.0000
Ward=48	0.01	-0.0000

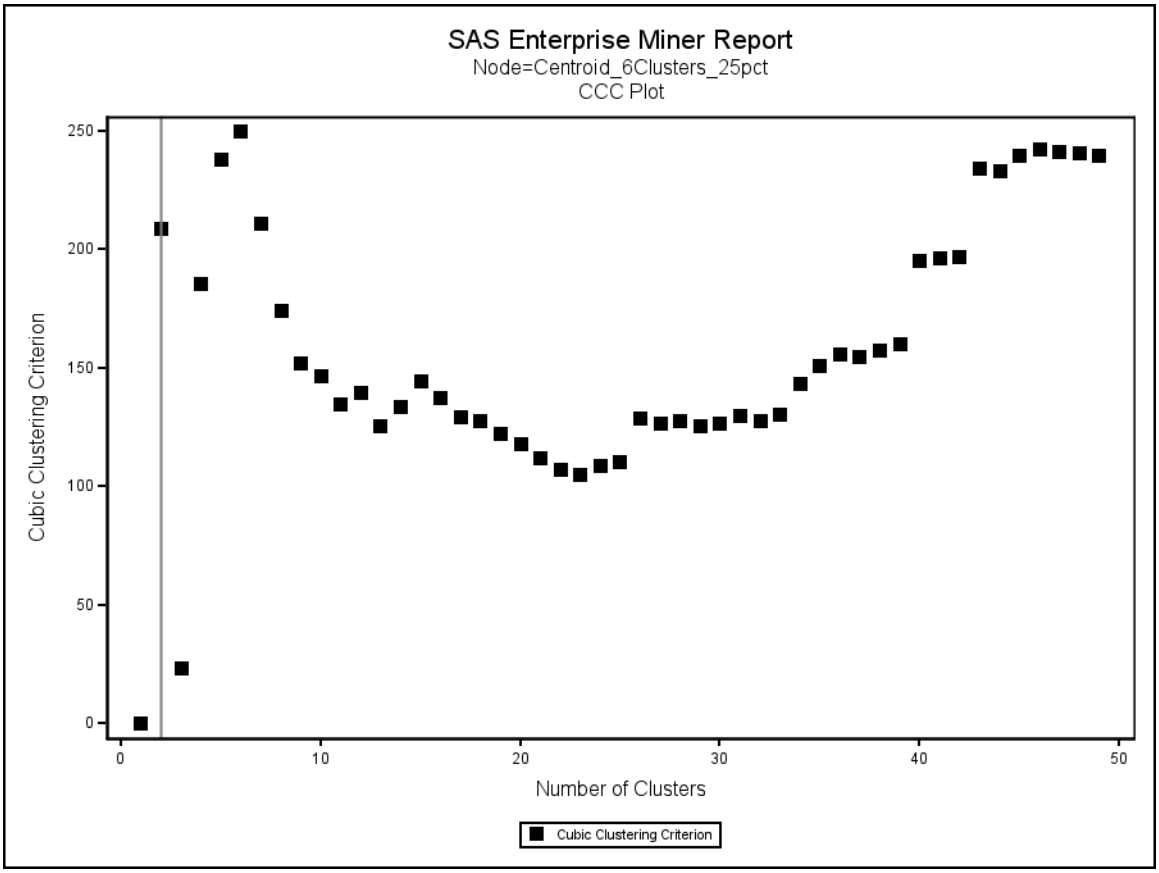
Variable	Segment 1	Segment 2
Ward=49	0.01	-0.0000
Ward=50	0.01	0.0000

SAS Enterprise Miner Report
Node=Centroid_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=Centroid_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=Centroid_6Clusters_25pct
CCC Plot
Cubic Clustering Criterion
250
200
150
100
50
0
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=Ward_6Clusters_25pct
Summary

Node id = Clus
Node label = Ward_6Clusters_25pct
Meta path = Ids => Smpl => Clus
Notes =

Node=Ward_6Clusters_25pct
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	WARD		ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	6	10	TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

Node=Ward_6Clusters_25pct
Variable Summary

Role	Level	Frequency Count	Name
INPUT	BINARY	2	Arrest Domestic
INPUT	INTERVAL	4	Hour Latitude Longitude Month
INPUT	NOMINAL	3	Community_Area District Ward
ID	INTERVAL	1	_dataobs_

Node=Ward_6Clusters_25pct
Mean Statistics

Variable	Segment 1	Segment 2	Segment 3
Clustering Criterion	0.2219	0.22	0.22
Maximum Relative Change in Cluster Seeds	0.0073	0.01	0.01
Improvement in Clustering Criterion	.	.	.
Frequency of Cluster	6.0000	105169.00	76811.00
Root-Mean-Square Standard Deviation	0.2091	0.22	0.22
Maximum Distance from Cluster Seed	3.3382	5.71	5.13
Nearest Cluster	2.0000	3.00	2.00
Distance to Nearest Cluster	36.4740	2.20	2.20
Hour	9.8333	13.06	12.92
Latitude	41.9809	41.90	41.76
Longitude	-87.8618	-87.70	-87.63
Month	8.0000	6.55	6.54

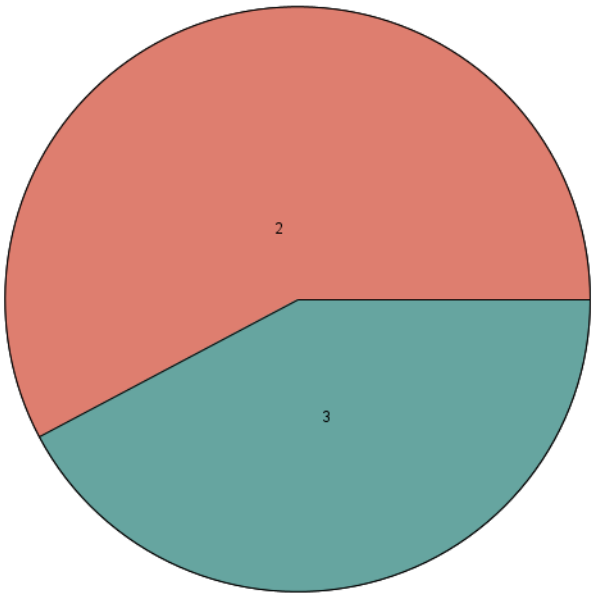
Variable	Segment 1	Segment 2	Segment 3
Arrest=0	0.8333	0.81	0.80
Arrest=1	0.1667	0.19	0.20
Community_Area=1	0.0000	0.03	0.00
Community_Area=2	0.0000	0.02	-0.00
Community_Area=3	-0.0000	0.02	-0.00
Community_Area=4	0.0000	0.01	0.00
Community_Area=5	-0.0000	0.01	-0.00
Community_Area=6	0.0000	0.04	-0.00
Community_Area=7	0.0000	0.03	-0.00
Community_Area=8	-0.0000	0.07	0.00
Community_Area=9	0.1667	0.00	-0.00
Community_Area=10	-0.0000	0.01	0.00
Community_Area=11	-0.0000	0.01	0.00
Community_Area=12	0.0000	0.00	0.00
Community_Area=13	0.0000	0.01	-0.00
Community_Area=14	0.0000	0.02	0.00
Community_Area=15	0.0000	0.02	0.00
Community_Area=16	0.0000	0.02	-0.00
Community_Area=17	0.0000	0.01	-0.00
Community_Area=18	-0.0000	0.00	0.00
Community_Area=19	-0.0000	0.03	0.00
Community_Area=20	0.0000	0.01	0.00
Community_Area=21	0.0000	0.01	-0.00
Community_Area=22	0.0000	0.03	0.00
Community_Area=23	-0.0000	0.05	0.00
Community_Area=24	0.1667	0.05	0.00
Community_Area=25	0.0000	0.10	-0.00
Community_Area=26	0.0000	0.04	0.00
Community_Area=27	0.0000	0.03	-0.00
Community_Area=28	0.0000	0.06	0.00
Community_Area=29	0.0000	0.06	-0.00
Community_Area=30	0.0000	0.03	0.00
Community_Area=31	0.0000	0.01	0.00
Community_Area=32	0.0000	0.06	0.00
Community_Area=33	0.0000	0.00	0.02
Community_Area=34	0.0000	0.00	0.01
Community_Area=35	-0.0000	-0.00	0.02
Community_Area=36	0.0000	0.00	0.01
Community_Area=37	0.0000	0.00	0.01
Community_Area=38	0.0000	-0.00	0.03
Community_Area=39	0.0000	0.00	0.01
Community_Area=40	-0.0000	-0.00	0.02
Community_Area=41	0.0000	-0.00	0.02
Community_Area=42	-0.0000	0.00	0.03
Community_Area=43	0.0000	-0.00	0.08
Community_Area=44	-0.0000	-0.00	0.06
Community_Area=45	0.0000	-0.00	0.01
Community_Area=46	0.0000	-0.00	0.04
Community_Area=47	-0.0000	0.00	0.00
Community_Area=48	0.0000	0.00	0.01
Community_Area=49	0.0000	0.00	0.06

Variable	Segment 1	Segment 2	Segment 3
Community_Area=50	0.0000	-0.00	0.01
Community_Area=51	0.0000	0.00	0.02
Community_Area=52	0.0000	-0.00	0.01
Community_Area=53	0.0000	0.00	0.04
Community_Area=54	0.0000	-0.00	0.01
Community_Area=55	0.0000	0.00	0.01
Community_Area=56	0.0000	0.01	0.00
Community_Area=57	-0.0000	0.01	0.00
Community_Area=58	0.0000	0.01	0.00
Community_Area=59	0.0000	0.00	0.00
Community_Area=60	0.0000	0.00	0.01
Community_Area=61	0.0000	0.00	0.04
Community_Area=62	0.0000	0.01	0.00
Community_Area=63	0.0000	0.01	0.01
Community_Area=64	0.0000	0.01	0.00
Community_Area=65	0.0000	0.01	0.00
Community_Area=66	0.0000	0.00	0.05
Community_Area=67	0.0000	0.00	0.06
Community_Area=68	0.0000	0.00	0.06
Community_Area=69	0.0000	-0.00	0.06
Community_Area=70	0.0000	0.00	0.02
Community_Area=71	-0.0000	-0.00	0.07
Community_Area=72	0.0000	-0.00	0.01
Community_Area=73	0.0000	-0.00	0.03
Community_Area=74	0.0000	-0.00	0.00
Community_Area=75	-0.0000	0.00	0.02
Community_Area=76	0.6667	0.01	-0.00
Community_Area=77	-0.0000	0.02	0.00
District=1	0.0000	0.07	0.03
District=2	0.0000	0.00	0.10
District=3	-0.0000	-0.00	0.11
District=4	0.0000	-0.00	0.13
District=5	0.0000	0.00	0.11
District=6	0.0000	-0.00	0.15
District=7	0.0000	0.00	0.13
District=8	-0.0000	0.04	0.08
District=9	-0.0000	0.02	0.07
District=10	0.0000	0.08	0.00
District=11	0.0000	0.13	-0.00
District=12	0.0000	0.08	0.00
District=14	0.0000	0.06	0.00
District=15	0.0000	0.07	-0.00
District=16	0.0000	0.06	0.00
District=17	0.0000	0.05	-0.00
District=18	-0.0000	0.09	0.00
District=19	0.0000	0.08	-0.00
District=20	0.0000	0.03	-0.00
District=22	0.0000	0.00	0.07
District=24	-0.0000	0.05	-0.00
District=25	0.0000	0.09	0.00
District=31	1.0000	-0.00	-0.00

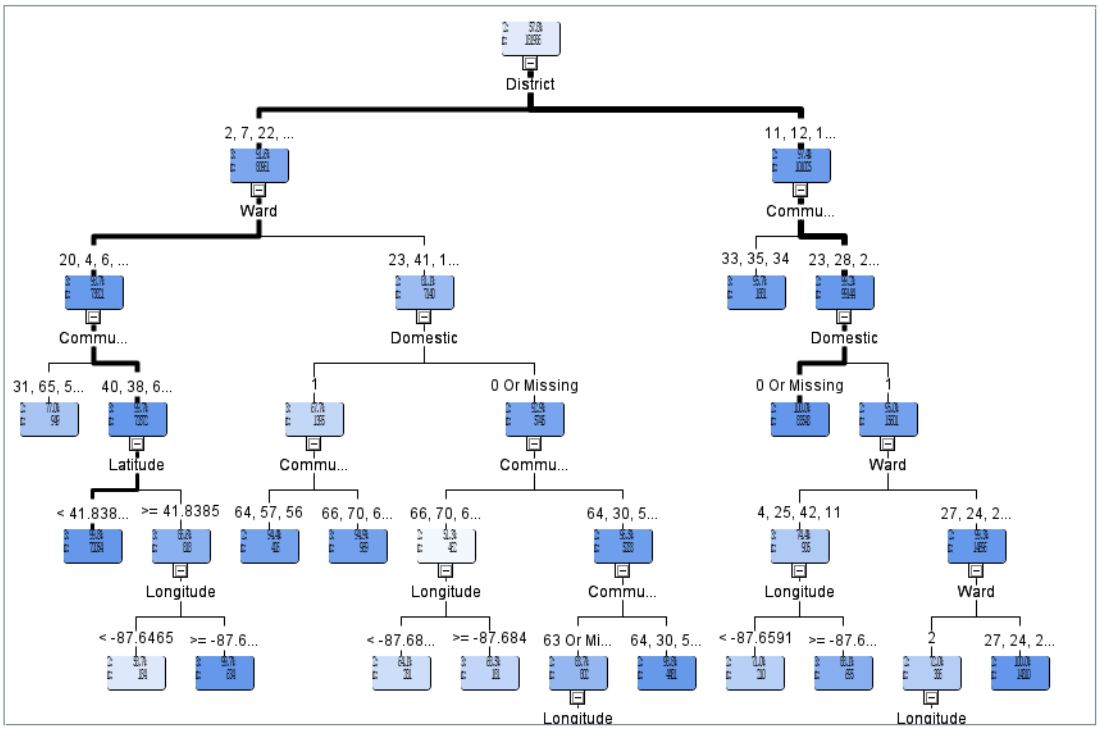
Variable	Segment 1	Segment 2	Segment 3
Domestic=0	1.0000	0.85	0.72
Domestic=1	-0.0000	0.15	0.28
Ward=1	0.1667	0.03	0.00
Ward=2	0.0000	0.04	0.01
Ward=3	0.0000	0.00	0.07
Ward=4	0.0000	0.01	0.05
Ward=5	-0.0000	0.00	0.06
Ward=6	0.0000	-0.00	0.09
Ward=7	0.0000	0.00	0.07
Ward=8	0.0000	0.00	0.07
Ward=9	0.0000	0.00	0.07
Ward=10	0.0000	-0.00	0.04
Ward=11	0.0000	0.01	0.02
Ward=12	-0.0000	0.02	0.00
Ward=13	0.0000	0.01	0.00
Ward=14	0.0000	0.02	0.01
Ward=15	0.0000	0.00	0.04
Ward=16	-0.0000	0.00	0.07
Ward=17	0.0000	-0.00	0.08
Ward=18	-0.0000	0.00	0.03
Ward=19	0.0000	-0.00	0.02
Ward=20	0.0000	-0.00	0.07
Ward=21	0.0000	0.00	0.07
Ward=22	-0.0000	0.02	0.00
Ward=23	0.0000	0.02	0.00
Ward=24	0.0000	0.08	0.00
Ward=25	-0.0000	0.02	0.01
Ward=26	-0.0000	0.03	-0.00
Ward=27	0.0000	0.08	0.00
Ward=28	0.0000	0.08	0.00
Ward=29	0.0000	0.04	-0.00
Ward=30	0.0000	0.02	0.00
Ward=31	-0.0000	0.02	-0.00
Ward=32	0.0000	0.02	-0.00
Ward=33	0.0000	0.02	0.00
Ward=34	0.0000	0.00	0.06
Ward=35	0.0000	0.02	0.00
Ward=36	0.0000	0.02	-0.00
Ward=37	0.0000	0.05	0.00
Ward=38	0.0000	0.01	0.00
Ward=39	0.0000	0.02	0.00
Ward=40	0.0000	0.02	0.00
Ward=41	0.8333	0.02	0.00
Ward=42	-0.0000	0.10	0.00
Ward=43	0.0000	0.02	0.00
Ward=44	0.0000	0.03	0.00
Ward=45	-0.0000	0.02	-0.00
Ward=46	0.0000	0.02	0.00
Ward=47	0.0000	0.02	-0.00
Ward=48	-0.0000	0.02	-0.00

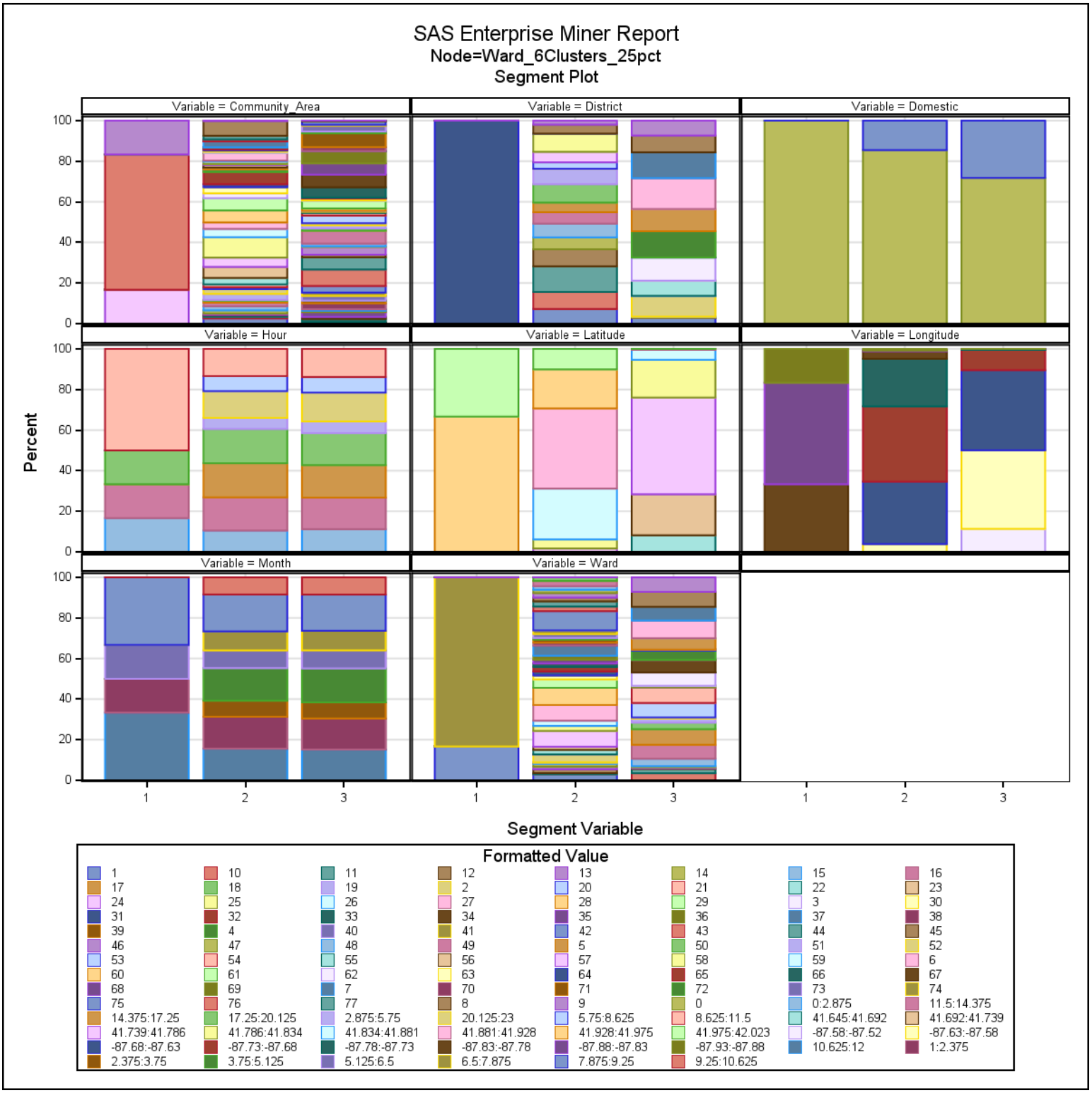
Variable	Segment 1	Segment 2	Segment 3
Ward=49	-0.0000	0.03	-0.00
Ward=50	0.0000	0.02	0.00

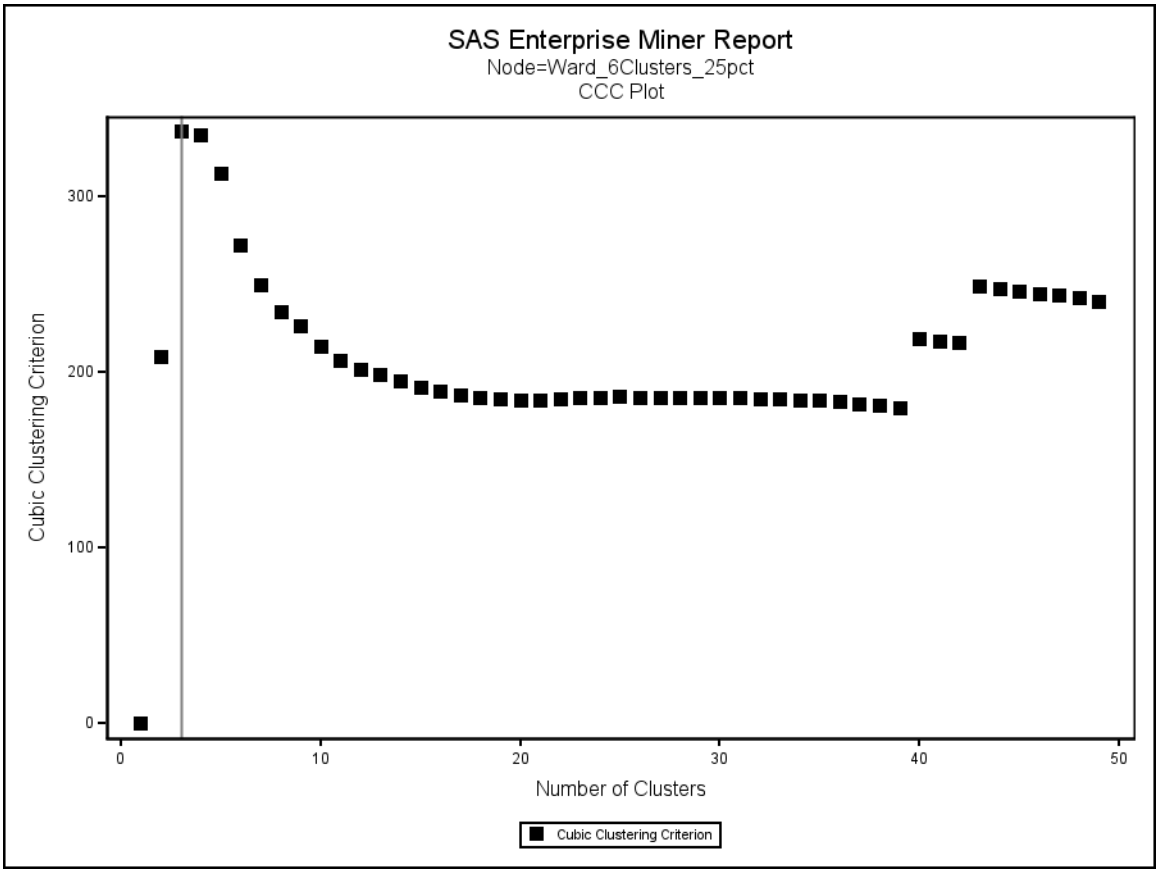
SAS Enterprise Miner Report
Node=Ward_6Clusters_25pct
Segment Size



SAS Enterprise Miner Report
Node=Ward_6Clusters_25pct
Tree Diagram




SAS Enterprise Miner Report
Node=Ward_6Clusters_25pct
Segment Plot
Variable = Community_Area
Variable = District
Variable = Domestic
Variable = Hour
Variable = Latitude
Variable = Longitude
Variable = Month
Variable = Ward
Percent
Segment Variable
Formatted Value
1
10
11
12
13
14
15
16
17
18
19
2
20
21
22
23
24
25
26
27
28
29
3
30
31
32
33
34
35
36
37
38
39
4
40
41
42
43
44
45
46
47
48
49
5
50
51
52
53
54
55
56
57
58
59
6
60
61
62
63
64
65
66
67
68
69
7
70
71
72
73
74
75
76
77
8
9
0
0:2.875
11.5:14.375
14.375:17.25
17.25:20.125
2.875:5.75
20.125:23
5.75:8.625
8.625:11.5
41.645:41.692
41.692:41.739
41.739:41.786
41.786:41.834
41.834:41.881
41.881:41.928
41.928:41.975
41.975:42.023
-87.58:-87.52
-87.63:-87.58
-87.68:-87.63
-87.73:-87.68
-87.78:-87.73
-87.83:-87.78
-87.88:-87.83
-87.93:-87.88
10.625:12
1:2.375
2.375:3.75
3.75:5.125
5.125:6.5
6.5:7.875
7.875:9.25
9.25:10.625



End of Report